# IRRATIONALITY OF $G \pm 5\pi^2/48$ AND IRRATIONALITY EXPONENTS

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ABSTRACT. Let $G$ be Catalan's constant and put $\lambda = G/2 - 5\pi^2/96$, so that $G - 5\pi^2/48 = 2\lambda$. We prove that $\lambda$ is irrational and that $|\lambda - a/b| > b^{-57.85}$ for every integer $a$ and every sufficiently large integer $b$; thus $\mu(\lambda) \le 57.85$. The same holds for the mirror constant $\lambda' = G/2 + 5\pi^2/96$, so that $G + 5\pi^2/48 = 2\lambda'$: $\lambda'$ is irrational and $\mu(\lambda') \le 57.85$ (the two constants satisfy $\lambda + \lambda' = G$, but the two measures do not combine to a measure for $G$). The proof uses the determinants $P_h(\lambda)$ of the $h \times h$ matrices $T_h = \lambda I + M_0$, which are (up to a factor $1/16$) the Löwner matrices of $\psi(z+1) - \psi(z+3/4)$ at the nodes $0, \dots, h-1$. The common denominator of $P_h$ satisfies $\log \operatorname{den} P_h \le (313/105 + o(1))h^2$; a contour representation of the Löwner matrix and Andréief's identity give $\log |P_h(\lambda)| \le -(3.1113\ldots - o(1))h^2$, the energy step using the conditionally positive definite kernel $-\frac{1}{2}\log(|z-w|^2 + \rho^2)$ and the numerical input being an interval-arithmetic certificate for a measure with 300 point masses; and an $\ell$-adic Schur-complement argument shows that $P_h$ cannot vanish at a given rational simultaneously for the two values $h = \ell - K$, $\ell - K - 1$ ($K = (\ell+1)/4$) attached to any prime $\ell \equiv 3 \pmod 4$; this rests on a closed-form evaluation (a product formula) of a family of Krylov determinants of Cauchy-type Gram matrices. For the measure we prove a node-removal bound on every principal minor of $T_h$, again with a certified numerical constant, which yields an exponentially wide window around $\lambda$ in which $P_h$ is small. The Lean 4 formalization proves the irrationality and $\mu(\lambda) \le 77.28$ (exponent $31682445459/409965000$; with the simpler choice of $h$), unconditionally; the determinant evaluation of §6 and the bound 57.85 are not formalized. The mirror theorem is formalized with the same weaker exponent (78 for $\lambda'$, 79 for $G + 5\pi^2/48$).

## 1. INTRODUCTION

OpenAI [10] recently proved that Catalan's constant

$$G = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = 0.91596\,55941\ldots$$

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is irrational, and formalized the proof in Lean. (An earlier claimed proof [13] was found to contain a gap [14].) Their argument, and the irrationality-measure results built on it [5, 2, 6], use a determinant of large size whose denominators and decay are balanced very closely.

In the present paper we study a different, much smaller, family of determinants, attached to the number

$$\lambda \;=\; \frac{G}{2} - \frac{5\pi^2}{96} \;=\; \sum_{k\ge 1}\Big(\frac{1}{16k^2} - \frac{1}{(4k-1)^2}\Big) \;=\; -0.05605\ldots$$

(see §2). Equivalently $G - 5\pi^2/48 = 2\lambda$. We also treat the "mirror" constant $\lambda' = G/2 + 5\pi^2/96$ (§9), for which $G + 5\pi^2/48 = 2\lambda'$ and $\lambda + \lambda' = G$.

**Theorem 1.1.** *The number $\lambda = G/2 - 5\pi^2/96$ is irrational. Equivalently, $G - 5\pi^2/48 \notin \mathbb{Q}$.*

**Theorem 1.2.** *For every integer $a$ and every sufficiently large integer $b$,*

$$\left|\lambda - \frac{a}{b}\right| > b^{-57.85}.$$

*In particular $\mu(\lambda) \le 57.85$ (the computed value is 57.841).*

A simpler choice of $h$ (without the determinant evaluation of §6) gives the weaker bound 77.13; see Remark 8.3. The Lean formalization proves, unconditionally, the irrationality and $\mu(\lambda) \le 77.28$ (exponent 31682445459/409965000; with the simpler choice of $h$; the value 77.28 rather than 77.13 comes from rounding the certified constants to short rationals); the determinant evaluation of §6 and the bound 57.85 are not formalized.

**Theorem 1.3** (the mirror constant)**.** *The number $\lambda' = G/2 + 5\pi^2/96$ is irrational, and for every integer $a$ and every sufficiently large integer $b$, $|\lambda' - a/b| > b^{-57.85}$; thus $\mu(\lambda') \le 57.85$. Consequently $G + 5\pi^2/48 \notin \mathbb{Q}$ and $\mu(G + 5\pi^2/48) \le 57.85$.*

The proof of Theorem 1.3 (§9) is the same architecture with the frame $\psi(z+1) - \psi(z+\frac{1}{4})$; the constants (content 313/105, the decay certificate, $c_{\mathrm{cert}}$) are identical, and only the $\ell$-adic part and some $o(1)$ bookkeeping change. Since $\lambda + \lambda' = G$ it is natural to ask whether Theorems 1.2 and 1.3 give a measure for $G$; they do not (Remark 9.7). The Lean formalization does not cover Theorem 1.3.

Here the *irrationality exponent* $\mu(\xi)$ of a real number $\xi$ is the infimum of the $\mu$ such that $|\xi - a/b| > b^{-\mu}$ for all integers $a$ and all sufficiently large $b$. We stress that Theorem 1.1 concerns one specific linear form in $1, G, \pi^2$; it is not a statement about $G$ alone, whose irrationality is OpenAI's theorem.

A search of the literature (including work of Rivoal and Zudilin and of Calegari, Dimitrov and Tang on linear forms involving $G$) did not turn up a previous proof of the irrationality of $G - 5\pi^2/48$ or of $G + 5\pi^2/48$.

1.1. **Method.** For $h \geq 1$ let $M_0$ be the explicit rational $h \times h$ matrix of §2 and $T_h = \lambda I + M_0$; thus $P_h(X) = \det(XI + M_0) \in \mathbb{Q}[X]$ and $P_h(\lambda) = \det T_h$.

(i) *Content* (§3). A prime-by-prime tropical (assignment) bound, with an explicit dual certificate, gives $\log \operatorname{den} P_h \leq (313/105 + o(1))h^2$.

(ii) *Decay* (§4). $T_h$ is $\frac{1}{16}$ times a Löwner matrix. Moving the Hermite contour integral to a two-ray contour produces finitely many signed atoms plus an integral against an exponentially small analytic weight, and Andréief's identity then bounds $|P_h(\lambda)| \leq 16^{-h} \det \mathcal{G}_h$ with $\mathcal{G}_h$ a positive definite Gram matrix. The energy of the resulting point configuration is bounded below with the kernel $k_\rho(z,w) = -\frac{1}{2}\log(|z-w|^2 + \rho^2)$, $\rho = 1/1024$, which is conditionally positive definite on finite sums (Frullani's integral and Gaussian sums), against a certificate measure $\sigma$ of 300 dyadic point masses; the resulting finite inequalities are checked in interval arithmetic and give $\det \mathcal{G}_h \leq \exp(-(3.1113 - o(1))h^2)$. The decay beats the content by the margin $3.1113790 - 313/105 > 0.1304$.

(iii) *Nonvanishing* (§5). For a prime $\ell \equiv 3 \pmod 4$, $\ell \geq 7$, and any rational $r$ with $\ell \nmid \operatorname{den} r$, one of $P_{\ell-1}(r), P_{\ell-2}(r), P_{(3\ell-1)/4}(r)$ is nonzero, because the $\ell$-adic Schur complement of $D(rI + M_0)D$ has a transparent Cauchy-matrix form. This gives irrationality with no information on the smallest eigenvalue of $T_h$.

(iv) *A determinant evaluation and the choice of $h$* (§6). Lemma CC says that two consecutive Gram-type matrices $N_K, N_{K+1}$ over $\mathbb{F}_\ell$ have no common eigenvalue. It is reduced to the non-vanishing modulo $\ell$ of a Krylov determinant $u(K,K) = \det[w, N_K w, \dots, N_K^{K-1} w]$, and Theorem A evaluates $|u|$ as a product of powers of integers $\leq 2K < \ell$ (zeros and poles in two shift parameters by Weyl's inequality and a cluster lemma, the constant by a recursion). Consequently, for any rational $r$ with $\ell \nmid \operatorname{den} r$, one of $P_{\ell-K}(r), P_{\ell-K-1}(r)$ is nonzero, and the two values of $h$ are within a factor $1 + O(1/\ell)$ of each other.

(v) *Measure* (§7–8). The proof of the decay bound extends to every principal minor of $T_h$: removing $j = \beta h$ nodes costs at most $e^{c\beta h^2}$, with $c = 2(m' - g_{\min}) \leq 6.8717994$ and $g_{\min}$ certified in interval arithmetic (Theorem 7.1). Expanding $\det(T_h + tI)$ in principal minors, this gives $|P_h(\lambda + t)| \leq e^{-(3.1113 - o(1))h^2}$ for all $|t| \leq e^{-ch}$. A rational $p/q$ very close to $\lambda$ lies in this window, and the arithmetic of the choice of $\ell$ and of $h \in \{\ell - K, \ell - K - 1\}$ yields $\mu(\lambda) \leq c\,(1/\eta + 3/4) < 57.85$, $\eta = 3.1113790\ldots - 313/105 = 0.1304266\ldots$ (the computed value is 57.841; with the simpler choice of $h$ of Theorem 5.4 one gets $c\,(4/(3\eta) + 1) < 77.13$, Remark 8.3; an earlier circle-smoothed energy step gave 79.35 with that choice, Remark 4.12).

1.2. **Status of the proofs.** The proofs of Theorems 1.1, 1.2 and 1.3 were derived by Claude (Anthropic) agents working with the author, and each part was re-derived and spot-checked numerically by separate Claude audit

agents; no human referee has checked them. Two steps are computer-assisted (interval-arithmetic certificates, §4.4 and §7.4). The Krylov determinant evaluation (Theorem A) and the choice of $h$ of §6 were derived and then audited by separate Claude agents (exact rational checks; see §10). The Lean formalization proves the irrationality and the weaker bound 77.13, conditional on the numerical certificate facts; Theorem A and the deduction of 57.85 are not yet formalized, and the mirror theorem, Theorem 1.3, is formalized with the same weaker exponent (78 for $\lambda'$, 79 for $G + 5\pi^2/48$). See §10 for an exact account.

1.3. **Notation.** $\psi$ is the digamma function, $H_n = \sum_{k\le n} 1/k$, $H_n^{(2)} = \sum_{k\le n} 1/k^2$, and $\vartheta(x)$, $\vartheta(x;4,a) = \sum_{p\le x,\, p\equiv a\,(4)} \log p$ are Chebyshev functions. For $P \in \mathbb{Q}[X]$, $\operatorname{den} P$ is the least positive integer $d$ with $dP \in \mathbb{Z}[X]$. Steps resting on numerical certificates are labelled "computer-assisted".

## 2. The matrices

2.1. **The number $\lambda$.** Since $\sum_{n\equiv1(4)} n^{-2} - \sum_{n\equiv3(4)} n^{-2} = G$ and the sum of both is $\pi^2/8$ (sums over $n \ge 1$),

$$\sum_{n\equiv3\,(4)} n^{-2} = \frac{\pi^2}{16} - \frac{G}{2}, \qquad \sum_{k\ge1} \frac{1}{16k^2} = \frac{\pi^2}{96},$$

so

$$\lambda := \sum_{k\ge1}\Big(\frac{1}{16k^2} - \frac{1}{(4k-1)^2}\Big) = \frac{\pi^2}{96} - \frac{\pi^2}{16} + \frac{G}{2} = \frac{G}{2} - \frac{5\pi^2}{96}. \tag{2.1}$$

2.2. **Löwner structure.** Let

$$F(z) = \psi(z+1) - \psi(z+\tfrac34) = \sum_{j\ge0}\Big[\frac{1}{z+j+\frac34} - \frac{1}{z+j+1}\Big],$$

meromorphic with simple poles at $-(j+\frac34)$ (residue $+1$) and $-(j+1)$ (residue $-1$), $j \ge 0$. For nodes $x_0,\dots,x_{h-1}$ the *Löwner matrix* of $F$ is $L_F[a][b] = (F(x_b) - F(x_a))/(x_b - x_a)$ for $a \ne b$ and $L_F[a][a] = F'(x_a)$. Take the nodes to be $x_a = a$, $a = 0,\dots,h-1$.

From $F(k) - F(k-1) = \frac{1}{k} - \frac{1}{k-1/4} = -\frac{1}{k(4k-1)}$ and the corresponding identity for $F'$,

$$L_F[a][b] = -\frac{1}{b-a}\sum_{a<k\le b} \frac{1}{k(4k-1)} \qquad (a<b), \tag{2.2}$$

$$L_F[a][a] = F'(0) + \sum_{k\le a}\Big(\frac{16}{(4k-1)^2} - \frac{1}{k^2}\Big), \tag{2.3}$$

and $F'(0) = \sum_{j\ge0}[(j+1)^{-2} - (j+\frac34)^{-2}] = \frac{\pi^2}{6} - 16\sum_{n\equiv3(4)} n^{-2} = 8G - \frac{5\pi^2}{6} = 16\lambda$.

**Definition 2.1.** Let $M_0 = M_0^{(h)}$ be the symmetric rational $h \times h$ matrix with, for $0 \le a < b < h$,

$$M_0[a][b] = -\frac{1}{4(b-a)} \sum_{k=a+1}^{b} \frac{1}{4k(4k-1)}, \qquad M_0[a][a] = -\sum_{k=1}^{a} \Big( \frac{1}{16k^2} - \frac{1}{(4k-1)^2} \Big),$$

and set $T_h = \lambda I + M_0$ and $P_h(X) = \det(XI + M_0) \in \mathbb{Q}[X]$, monic of degree $h$.

**Proposition 2.2.** *$T_h = \frac{1}{16} L_F$ at the nodes $0, \dots, h-1$. In particular $P_h(\lambda) = 16^{-h} \det L_F$, and $P_h(\lambda + t) = \det(T_h + tI)$ for $t \in \mathbb{R}$.*

*Proof.* Compare (2.2), (2.3) with the definition, using $F'(0) = 16\lambda$ and $\frac{1}{16k^2} - \frac{1}{(4k-1)^2} = \frac{1}{16}\big(\frac{1}{k^2} - \frac{16}{(4k-1)^2}\big)$. □

*Remark* 2.3. The coefficients of $P_h$ are sums of principal minors of $M_0$, whose entries are harmonic-type partial sums; there is no factorial closed form. Also $L_F$ is not a Gram matrix of a positive measure, since the "Stieltjes measure" of $F$ consists of the signed atoms $+1$ at $-(j + \frac{3}{4})$ and $-1$ at $-(j+1)$; this is the reason for the contour step in §4.

**Lemma 2.4** (integrality)**.** *Let $Q_h = \mathrm{den}(P_h) \cdot P_h \in \mathbb{Z}[X]$. If $a/b$ is in lowest terms and $Q_h(a/b) \neq 0$, then $|Q_h(a/b)| \ge b^{-h}$.*

*Proof.* $Q_h$ has degree $h$ and integer coefficients, so $b^h Q_h(a/b) \in \mathbb{Z} \setminus \{0\}$. □

## 3. The denominator bound

**Theorem 3.1.** $\log \mathrm{den}\, P_h \le (\frac{313}{105} + o(1))h^2$ *as* $h \to \infty$.

We write $\Lambda = 313/105 = 2.98095\ldots$. The proof is prime by prime. For a prime $p$ put $W_{ab} = \max(0, -v_p(M_0[a][b]))$.

### 3.1. Tropical bound.

**Lemma 3.2.** *Let $\tau_p$ be the maximum of $\sum_a W_{a,\sigma(a)}$ over permutations $\sigma$ of $\{0, \dots, h-1\}$. Then $v_p(\mathrm{den}\, P_h) \le \tau_p$. If $u_a, v_b \ge 0$ satisfy $W_{ab} \le u_a + v_b$ for all $a, b$, then $\tau_p \le \sum_a u_a + \sum_b v_b$.*

*Proof.* The coefficient of $X^{h-j}$ in $\det(XI + M_0)$ is a sum of $j \times j$ principal minors, each a signed sum of products $\prod_{a \in S} M_0[a][\sigma(a)]$ over permutations $\sigma$ of a $j$-set $S$. Each product has $-v_p \le \sum_{a \in S} W_{a\sigma(a)}$, and since $W \ge 0$, extending $\sigma$ by the identity off $S$ shows this is at most $\tau_p$. For the second statement, $\sum_a W_{a\sigma(a)} \le \sum_a (u_a + v_{\sigma(a)})$. □

### 3.2. Valuation majorant.

Let $p$ be odd and $\nu(k) = v_p(k(4k-1)) = \max(v_p(k), v_p(4k-1))$ (the factors $k$ and $4k-1$ are coprime). Let $\mu(a) = \max_{1 \le k \le a} \nu(k)$. By the ultrametric inequality applied to the formulas of the Definition,

$$W_{ab} \le v_p(b-a) + \max_{a < k \le b} \nu(k) \ (a < b), \qquad W_{aa} \le 2\mu(a). \tag{3.1}$$

For $q = p^e$ set

$$\operatorname{div}_q(a,b) = \mathbf{1}[q \mid b-a,\ a \neq b],$$
$$\operatorname{supp}_q(a,b) = \mathbf{1}[\exists k \in (\min(a,b), \max(a,b)] :\ q \mid k(4k-1)]\ (a \neq b), \qquad \operatorname{supp}_q(a,a) = 2 \cdot \mathbf{1}[\mu(a) \geq e].$$

Because $v_p(n) = \sum_{e \geq 1} \mathbf{1}[p^e \mid n]$ and $\max_k \nu(k) = \sum_e \mathbf{1}[\exists k :\ p^e \mid k(4k-1)]$, (3.1) says $W \leq \sum_{e \geq 1} w^{(e)}$ entrywise with $w^{(e)} = \operatorname{div}_{p^e} + \operatorname{supp}_{p^e} \in \{0,1,2\}$. Since a maximum of a sum is at most the sum of maxima, $\tau_p \leq \sum_e \tau(w^{(e)})$.

3.3. **Per-level dual certificate.** Let $q = p^e$ be an odd prime power. The least $k \geq 1$ with $q \mid k(4k-1)$ is $k_q = (j_q q + 1)/4$, where $j_q = 1$ if $q \equiv 3 \pmod 4$ and $j_q = 3$ if $q \equiv 1 \pmod 4$ (indeed $4k - 1 = jq$ forces $jq \equiv 3 \pmod 4$, and the branch $q \mid k$ gives $k \geq q > k_q$). Hence $\operatorname{supp}_q(a,b) \neq 0$ forces $\max(a,b) \geq k_q$. Put $n_q = \min(h, k_q)$ and define potentials

$$u_a = v_a = \begin{cases} 1 & a \geq k_q \text{ (“high”)}, \\ \mathbf{1}[a \leq h - 1 - q] & a < k_q \text{ (“low”)}. \end{cases}$$

**Lemma 3.3.** *$w^{(e)}_{ab} \leq u_a + v_b$ for all $a, b$. Consequently*

$$\tau(w^{(e)}) \leq \delta(q) := \min\bigl(2h,\ 2(h - n_q) + 2\min(n_q, (h-q)_+)\bigr),$$

*and $\delta(q) = 0$ if $q \geq h$ and $k_q \geq h$; in particular $\tau_p = 0$ for $p > 4h$.*

*Proof.* Note $k_q < q$. *Both low:* $|b - a| < k_q < q$ so div $= 0$, and $(a,b] \subset [1, k_q - 1]$ so supp $= 0$ (diagonal likewise). *Both high:* $w \leq 2 = u_a + v_b$. *$a$ low, $b$ high:* supp $\leq 1 = v_b$; if div $= 1$ then $b \geq a + q$, $b \leq h - 1$, so $a \leq h - 1 - q$ and $u_a = 1$; the case $a$ high, $b$ low is symmetric. The diagonal is 0 at low and at most 2 at high indices. Now apply Lemma 3.2: $\sum u + \sum v = 2\#\{\text{high}\} + 2\#\{a < n_q :\ a \leq h - 1 - q\}$, and the trivial potentials $u = v \equiv 1$ give $2h$. □

By Lemma 3.2, $v_p(\operatorname{den} P_h) \leq \sum_{e \geq 1} \delta(p^e)$ for odd $p$.

3.4. **Summation. Prime 2.** Every entry of $M_0$ has $-v_2 \leq 4 + 2\log_2 h$ (from the factors $1/(4(b-a))$, $1/(4k(4k-1))$, $1/(16k^2)$), so the prime 2 contributes $O(h \log h)$ to $\log \operatorname{den} P_h$.

**Levels $e \geq 2$.** Here $\delta \leq 2h$, so these contribute at most $2h\,(\sum_{p^e \leq 4h, e \geq 2} \log p) = 2h(\psi_{\mathrm{Ch}}(4h) - \vartheta(4h)) = O(h^{3/2})$ (Chebyshev: $\psi_{\mathrm{Ch}}(x) - \vartheta(x) = O(\sqrt{x})$).

**Level $e = 1$.** For $q = p$ a direct computation from Lemma 3.3 (using $|k_q - j_q q/4| \leq 1/4$ and that $\delta$ is 2-Lipschitz in $n_q$) gives $\delta(p) \leq h\, g_j(p/h) + \frac{1}{2}$, $j = j_p$, where

$$g_j(\theta) = \begin{cases} 2 & 0 \leq \theta \leq \frac{4}{4+j}, \\ 4 - \frac{(4+j)\theta}{2} & \frac{4}{4+j} \leq \theta \leq 1, \\ 2 - \frac{j\theta}{2} & 1 \leq \theta \leq \frac{4}{j}, \\ 0 & \theta \geq \frac{4}{j}. \end{cases}$$

Indeed for $\theta \leq 4/(4+j)$ one has $n_q \leq h - q$ and $\delta = 2h$; for $4/(4+j) \leq \theta \leq 1$, $\delta = 2(h - n_q) + 2(h - q)$; for $\theta \geq 1$, $\delta = 2(h - n_q)$. The function $g_j$ is

continuous, non-increasing and piecewise linear, with $\int_0^\infty g_j = \frac{4}{j} + \frac{j}{4+j}$, that is

$$\int g_1 = \frac{21}{5} \ (p \equiv 3 \bmod 4), \qquad \int g_3 = \frac{37}{21} \ (p \equiv 1 \bmod 4).$$

By partial summation, since $-g_j' \geq 0$ is supported on $\theta \geq 4/7$,

$$\frac{1}{h} \sum_{p \equiv a\,(4)} \log p \; g_j(p/h) = \frac{1}{h} \int (-g_j'(\theta)) \, \vartheta(\theta h; 4, a) \, d\theta \leq \frac{1+\varepsilon_h}{2} \int g_j,$$

where $\varepsilon_h = \sup_{4h/7 \leq x \leq 4h} (2\vartheta(x;4,a)/x - 1) \to 0$ by the prime number theorem in arithmetic progressions mod 4 (de la Vallée Poussin; see e.g. [3, Ch. 20] or [9, Cor. 11.19]); only an upper bound for $\vartheta(x;4,a)$ is needed because the weights are non-negative. Collecting terms,

$$\frac{\log \operatorname{den} P_h}{h^2} \leq (1+\varepsilon_h)\Big(\frac{1}{2} \cdot \frac{21}{5} + \frac{1}{2} \cdot \frac{37}{21}\Big) + O\Big(\frac{1}{\sqrt{h}}\Big) = (1+\varepsilon_h)\frac{313}{105} + o(1).$$

This proves Theorem 3.1. □

*Remark* 3.4 (computer-assisted supplement; not used). Evaluating the per-$h$ bound $\sum_q \delta(q) \log p$ (plus a bound for $p = 2$) exactly for every $h \leq 2 \cdot 10^5$ gives $\log \operatorname{den} P_h \leq 2.98635\, h^2$, with maximum at $h = 1844$ (script `scan.py`). This is not needed for the theorems. The dual certificate is also essentially sharp: for odd primes and $h = 64, \ldots, 256$ the model value equals the exact tropical optimum except at $p = 3$.

## 4. The decay bound

4.1. **A contour representation of $L_F$.** Hermite's formula gives, for any contour $C$ enclosing the nodes and no pole of $F$,

$$L_F = \frac{1}{2\pi i} \oint_C F(z)\, v(z) v(z)^\top dz, \qquad v_a(z) = \frac{1}{z-a}$$

(the residue at $z = a \neq b$ gives $(F(a) - F(b))/(a-b)$, the double pole gives $F'(a)$). By the reflection formulas $\psi(1+z) = \psi(-z) - \pi \cot \pi z$ and $\psi(\frac{3}{4}+z) = \psi(\frac{1}{4}-z) + \pi \cot(\frac{\pi}{4} - \pi z)$,

(4.1)

$$F = \tilde{F} + E, \qquad \tilde{F}(z) = \psi(-z) - \psi(\tfrac{1}{4} - z), \quad E(z) = -\pi\big[\cot \pi z + \cot(\tfrac{\pi}{4} - \pi z)\big] = \frac{-\pi \sin(\pi/4)}{\sin \pi z \, \sin(\pi/4 - \pi z)}.$$

Here $\tilde{F}$ is analytic in $\operatorname{Re} z < 0$ and $O(1/|z|)$ there, while $E$ is 1-periodic, has poles only on the real axis, and $|E(x+iy)| \leq Ce^{-2\pi|y|}$ for $|y| \geq 1$.

Fix $r_0 = 7/50$, $N_0 = \lfloor r_0 h \rfloor$, $x^* = -N_0 - \frac{3}{8}$ (which lies in the gap $(-N_0 - \frac{3}{4}, -N_0)$ between poles of $F$). Let $\Gamma_h$ be the two rays $x^* + u(-1 \pm 3i)$, $u \geq 0$ (the *rational direction* $(-1, \pm 3)$, at angle $\pi - \arctan 3 = 0.6024\pi$ from the positive axis), traversed downward, and let $\mathcal{A} = \{j + \frac{3}{4},\ j+1 :\ 0 \leq j < N_0\}$, $|\mathcal{A}| = 2N_0$. For $\mu \in \mathcal{A}$ let $c_\mu = +1$ for $\mu = j + \frac{3}{4}$ and $-1$ for $\mu = j+1$, and $u_\mu(a) = 1/(a+\mu)$. The direction enters only through three facts: the left wedge bounded by $\Gamma_h$ has half-angle $\arctan 3 < \pi/2$, so $\operatorname{Re} z \leq x^* < 0$ on it;

$E$ and $x^* \equiv 5/8 \pmod 1$ are periodic; and on the part of $\Gamma_h$ with $|\operatorname{Im} z| < 1$ (that is, $u < \frac{1}{3}$) one has $\operatorname{Re} z \in (x^* - \frac{1}{3}, x^*] = (-N_0 - 0.7083\ldots, -N_0 - \frac{3}{8}]$, strictly inside the gap between the poles $-N_0 - \frac{3}{4}$ and $-N_0$.

**Proposition 4.1** (contour identity)**.** *For every $h$,*

$$L_F = -\sum_{\mu\in\mathcal{A}} c_\mu\, u_\mu u_\mu^\top + \frac{1}{2\pi i}\int_{\Gamma_h} E(z)\, v(z)v(z)^\top dz.$$

*Proof.* Move the contour in Hermite's formula from around the nodes to $\Gamma_h$, closing on the right where the integrand is $O(|z|^{-3})$; this picks up the residues of $F$ at the $2N_0$ poles $-\mu$, $\mu \in \mathcal{A}$, which lie to the right of $x^*$, each contributing $\pm c_\mu u_\mu u_\mu^\top$ according to orientation. Then split $F = \tilde{F} + E$; the $\tilde{F}$ part integrates to 0 over $\Gamma_h$, by closing in the left wedge where $\tilde{F}vv^\top$ is analytic and $O(|z|^{-3})$. □

(The identity, with the stated orientation and signs, was checked entrywise to $10^{-17}$ with 25-digit arithmetic at $N_0 = 2$ and $N_0 = 7$, for the direction $(-1, 3)$ and for the earlier angle $3\pi/5$.)

4.2. **Absolute-value Andréief bound.** Let $|m| = \sum_{\mu\in\mathcal{A}} \delta_{-\mu} + \frac{1}{2\pi}|E(z)||dz|$ on $\Gamma_h$, and

$$\mathcal{G}_h = \sum_{\mu\in\mathcal{A}} u_\mu u_\mu^\top + \frac{1}{2\pi}\int_{\Gamma_h} |E|\, v\, v^*|dz|,$$

a Hermitian positive definite matrix (the $v_a$ are linearly independent functions).

**Proposition 4.2.** $|\det L_F| \le \det \mathcal{G}_h$*; hence* $|P_h(\lambda)| \le 16^{-h} \det \mathcal{G}_h$*. More generally, for real $t$,* $|P_h(\lambda + t)| \le \det(\mathcal{G}_h/16 + |t|I)$*. Moreover*

$$\det \mathcal{G}_h = \frac{1}{h!}\int \prod_{i<j} |z_i - z_j|^2 \prod_{a<b} (b-a)^2 \prod_{i,a} |z_i - a|^{-2} \prod_i d|m|(z_i).$$

*Proof.* By Proposition 4.1, $L_F[a][b] = \int v_a v_b\, dm$ for a complex measure $m$ with total variation $|m|$. Andréief's identity [4] gives $\det L_F = \frac{1}{h!}\int \det[v_a(z_i)]^2 \prod dm(z_i)$, so $|\det L_F| \le \frac{1}{h!}\int |\det[v_a(z_i)]|^2 \prod d|m|(z_i) = \det \int v_a \overline{v_b}\, d|m| = \det \mathcal{G}_h$, and the displayed formula is Cauchy's determinant $\det[1/(z_i - a)] = \pm \prod_{i<j}(z_i - z_j) \prod_{a<b}(b-a) / \prod (z_i - a)$. For the last assertion, $T_h + tI$ is the Gram integral $\int \Phi_a \Phi_b\, dM$ on the disjoint union of $\Gamma_h \cup (-\mathcal{A})$ and the node set, with $\Phi_a = v_a/4$ on the first part and $\Phi_a = \mathbf{1}_{\{a\}}$ on the second, and $M = m \sqcup t \cdot \#$; the same argument bounds $|\det|$ by $\det(\mathcal{G}_h/16 + |t|I)$. □

4.3. **The kernel $k_\rho$.** Fix $\rho = 1/1024$ and put

$$k_\rho(z, w) := -\tfrac{1}{2}\log(|z - w|^2 + \rho^2) \qquad (z, w \in \mathbb{C}), \qquad k_\rho(z, z) = -\log \rho.$$

The energy step below uses $k_\rho$ in place of the logarithmic kernel. Its virtue is that the conditional positive definiteness it needs is a statement about *finite*

sums and is proved by three elementary integrals (no Fourier transform and no measure on $\mathbb{C}$), and that no regularity constant for the potential of the certificate measure is ever needed.

**Lemma 4.3** (comparison)**.** *For $z \neq w$, $k_\rho(z,w) \leq -\log|z-w|$.*

*Proof.* $|z-w|^2 < |z-w|^2 + \rho^2$ and log is increasing. □

**Lemma 4.4** (Frullani form of the logarithm)**.** *For $a > 0$, $\log a = \displaystyle\int_0^\infty \frac{e^{-t} - e^{-at}}{t}\,dt$, and the integrand is integrable on $(0,\infty)$.*

*Proof.* For $t > 0$, $(e^{-t} - e^{-at})/t = \int_1^a e^{-tu}\,du$ (oriented integral); the integrand is bounded by $|a-1|$ near 0 and by $2e^{-\min(1,a)t}$ on $[1,\infty)$. If $a \geq 1$ the function $(t,u) \mapsto e^{-tu}$ on $(0,\infty)\times[1,a]$ is non-negative and $\int_1^a (\int_0^\infty e^{-tu}\,dt)\,du = \int_1^a du/u = \log a < \infty$, so Tonelli lets us exchange the integrals. If $0 < a < 1$, interchange with reversed orientation (the integrand is then $-\int_a^1 e^{-tu}du$). □

**Lemma 4.5** (finite Gaussian sums)**.** *For $t \geq 0$, finitely many points $p_\alpha = (x_\alpha, y_\alpha) \in \mathbb{R}^2$ and reals $w_\alpha$, $\sum_{\alpha,\beta} w_\alpha w_\beta e^{-t|p_\alpha - p_\beta|^2} \geq 0$.*

*Proof.* Put $v_\alpha = w_\alpha e^{-t|p_\alpha|^2}$; the sum is $\sum v_\alpha v_\beta e^{2t\langle p_\alpha, p_\beta\rangle} = \sum_{n\geq 0} \frac{(2t)^n}{n!} \sum_{\alpha,\beta} v_\alpha v_\beta \langle p_\alpha, p_\beta\rangle^n$. By the binomial theorem $\langle p, p'\rangle^n = \sum_{k=0}^n \binom{n}{k} (x^k y^{n-k})(x'^k y'^{n-k})$, hence

$$\sum_{\alpha,\beta} v_\alpha v_\beta \langle p_\alpha, p_\beta\rangle^n = \sum_{k=0}^n \binom{n}{k} \Big(\sum_\alpha v_\alpha x_\alpha^k y_\alpha^{n-k}\Big)^2 \geq 0,$$

a finite identity. A finite sum of convergent series is the series of the finite sums, and a series of non-negative terms is non-negative. □

**Lemma 4.6** (conditional positive definiteness)**.** *Let $p_1, \dots, p_N \in \mathbb{C}$ (repetitions allowed) and $q_1, \dots, q_N \in \mathbb{R}$ with $\sum q_\alpha = 0$. Then $\sum_{\alpha,\beta} q_\alpha q_\beta\, k_\rho(p_\alpha, p_\beta) \geq 0$.*

*Proof.* Put $a_{\alpha\beta} = |p_\alpha - p_\beta|^2 + \rho^2 > 0$. By Lemma 4.4 and linearity, $\sum q_\alpha q_\beta(-\log a_{\alpha\beta}) = \int_0^\infty \sum_{\alpha,\beta} q_\alpha q_\beta \frac{e^{-ta_{\alpha\beta}} - e^{-t}}{t}\,dt$. The $e^{-t}$ part is $(\sum q_\alpha)^2 e^{-t}/t = 0$ pointwise. What remains is $\frac{e^{-t\rho^2}}{t} \sum q_\alpha q_\beta e^{-t|p_\alpha - p_\beta|^2} \geq 0$ by Lemma 4.5. Multiply by $\frac{1}{2}$. □

For a finite set $X \subset \mathbb{C}$ with $|X| = h$ and a certificate measure $\sigma = \sum_{k\leq n_\sigma} m_k \delta_{c_k}$ ($m_k > 0$, $\sum m_k = 1$) put

$$U_\rho^\sigma(z) = \sum_k m_k\, k_\rho(z, c_k), \qquad I_\rho(\sigma) = \sum_{k,l} m_k m_l\, k_\rho(c_k, c_l).$$

**Corollary 4.7.** $\displaystyle\frac{1}{h^2} \sum_{x\neq y\in X} k_\rho(x,y) \;\geq\; \frac{2}{h} \sum_{x\in X} U_\rho^\sigma(x) - I_\rho(\sigma) + \frac{\log\rho}{h}.$

*Proof.* Apply Lemma 4.6 to the list $(X, c_1, \dots, c_{n_\sigma})$ with weights $(\frac{1}{h}, \dots, \frac{1}{h}, -m_1, \dots, -m_{n_\sigma})$, which sum to 0. Expanding, $\frac{1}{h^2}\sum_{x,y\in X} k_\rho(x,y) - \frac{2}{h}\sum_x U^\sigma_\rho(x) + I_\rho(\sigma) \geq 0$, and the diagonal terms $x = y$ contribute $\frac{1}{h^2}\cdot h \cdot (-\log\rho)$ to the left, which we move across. □

4.4. **The energy functional and the certificate.** Scale $z = hs$ and put $\tau$ = Lebesgue measure on $[0,1]$ and $U^\tau(s) = -\int_0^1 \log|s-t|\,dt$, so that $I(\tau) = 3/2$. For $\operatorname{Re} z < 0$ and $y = \operatorname{Im} z \geq 0$ one has the closed form

$$U^\tau(x+iy) = 1 - x\log|z| - (1-x)\log|1-z| - y\arctan\frac{y}{x^2+y^2-x},$$

symmetric in $y \mapsto -y$; on the real line $U^\tau(t) = 1 - t\log|t| - (1-t)\log|1-t|$. Write $\Gamma$ for the rays $\{-r_0 + u(-1 \pm 3i) : u \geq 0\}$ and $\kappa = 10^{-3}$. For a fixed *certificate measure* $\sigma$ define

$$g(s) = U^\sigma_\rho(s) - U^\tau(s) + (1-\kappa)\pi|\operatorname{Im} s|.$$

The only requirement on $\sigma$ is that its quantities are computed rigorously; its relation to an optimizer is irrelevant.

**The measure $\sigma$.** $\sigma = \sum_{k=1}^{n_\sigma} m_k \delta_{c_k}$ is a sum of $n_\sigma = 300$ point masses (84 real points and 108 conjugate pairs, so $\sigma$ and $g$ are conjugation-symmetric), with dyadic data: coordinates of $c_k$ of the form $X/2^{24}$, $Y/2^{24}$ with integers $X, Y$ of at most 23 bits, and masses $n_k/2^{32}$ with $\sum m_k = 1$ *exactly*. The data are in the file `sigma300.json`. Everything below is a finite computation with these rationals; in particular there are no circles, no Lipschitz constant and no continuum energy $I(\sigma)$.

**What is certified.** Put $\omega = 2^{-30}$ (a box half-width absorbing the horizontal shift of $\Gamma_h/h$) and $u_{\max} = \frac{3}{2}$.

(R) A cover of $u \in [0, u_{\max}]$ by intervals $[u_a, u_b]$ such that $g \geq m'$ on each box $[-r_0 - u_b - \omega, -r_0 - u_a + \omega] \times [3u_a, 3u_b]$, together with the tail bound $T(u_{\max}) \geq m'$: for $u \geq u_{\max}$, $|z| \leq r_0 + \omega + u\sqrt{10}$, $U^\sigma_\rho(z) \geq -\frac{1}{2}\log((|z| + R_\sigma)^2 + \rho^2)$ with $R_\sigma = \max|c_k|$, $U^\tau(z) \leq -\log(|z| - 1)$, and the field term is $\geq (1-\kappa)3\pi u$; the resulting lower bound $T(u)$ is increasing.

(S) A partition of $[-r_0, 0]$ into cells $C_\ell = [a_\ell, b_\ell]$ with $g \geq \gamma_\ell$ on $C_\ell$. Put $S_{\text{cert}} := 2\sum_\ell (b_\ell - a_\ell)\min(\gamma_\ell - m', 0)$.

(L) A cover of $[0,1]$ (in fact of $[-\frac{1}{64}, 1]$) by cells on which $U^\sigma_\rho - U^\tau \geq g_{\min}$.

Then put

$$D_{\text{cert}} := 2(m' + S_{\text{cert}}) - I_\rho(\sigma) + \tfrac{3}{2}, \qquad c := 2(m' - g_{\min}).$$

On a box or cell the lower bound for $U^\sigma_\rho$ is $-\frac{1}{2}\sum_k m_k \log(d_{x,k}^2 + d_{y,k}^2 + \rho^2)$ with $d_{x,k}, d_{y,k}$ the largest horizontal and vertical distances from $c_k$ to the box, which makes every logarithm argument as large as possible on the box (the two coordinates decouple, so the maximum is attained); $U^\tau$ is bounded above on ray boxes by ball arithmetic on the closed form, on segment cells by its value at the right end (it is increasing on $(-\infty, 0]$), and on cells of

$[0,1]$ by monotonicity on each side of $\frac{1}{2}$ (the supremum is $U^\tau(b)$ if $b \le \frac{1}{2}$, $U^\tau(a)$ if $a \ge \frac{1}{2}$, and $1+\log 2$ otherwise).

**Certified numbers (computer-assisted).** The script `kcert.py` (Arb [1], 96-bit precision; ray boxes bisected adaptively) was run on `sigma300.json` with the following results; $S_{\mathrm{cert}}$ is assembled from exact rationals, and all other entries are lower or upper bounds as indicated.

| quantity | certified value | cells |
|---|---|---|
| $I_\rho(\sigma)$ (exact finite sum) | $\le 2.1107199269$ | $\binom{n_\sigma+1}{2}$ terms |
| $m'$: inf $g$ over the ray boxes, $u \in [0, \frac{3}{2}]$ | $\ge 1.9718000062$ | 26,977 boxes |
| tail $T(\frac{3}{2})$ | $\ge 13.735$ (analytic) | – |
| $S_{\mathrm{cert}}$ | $\ge -0.1107505186$ | 14,000 cells |
| $g_{\min}$ on $[-\frac{1}{64}, 1]$ | $\ge -1.4640996549$ | 4,423 cells |
| $D_{\mathrm{cert}}$ | $\ge 3.1113790$ | |
| $c = 2(m' - g_{\min})$ | $\le 6.8717994$ | |
| $\eta = D_{\mathrm{cert}} - \Lambda$ | $\ge 0.1304266$ | |
| $c\,(1/\eta + 3/4)$ (with Lemma 5.5) | $\le$ 57.85 (computed: 57.841) | |
| $c\,(1/\eta + 3/2)$ (without Lemma 5.5) | $<$ 63 (computed: 62.995) | |
| $c\,(4/(3\eta)+1)$ (simpler $h$, Remark 8.3) | $\le 77.1217$ (unrounded: 77.1212) | |

(The rational data of the certificate were also re-derived in exact arithmetic by `mkdata.py`, with the masses rounded down to $2^{-32}$; this gave $D \ge 3.11137905$ and $c \le 6.87179932$.) The certificate is valid for $h \ge 1/\omega = 2^{30}$. It is a computation about the fixed function $g$ and does not depend on $h$.

### 4.5. **The decay theorem.**

**Lemma 4.8** (pointwise energy inequality $A_\rho$)**.** *Let $w_\kappa = |E|^{1-\kappa}$ on $\Gamma_h$ and $w_\kappa = 1$ at the atoms. For every configuration $z_1, \dots, z_h \in \Gamma_h \cup (-\mathcal{A})$ of pairwise distinct points (otherwise the Cauchy determinant vanishes), writing $s_i = z_i/h$,*

$$-\log\Big(\prod_{i<j}|z_i-z_j|^2 \prod_{a<b}(b-a)^2 \prod_{i,a}|z_i-a|^{-2}\prod w_\kappa(z_i)\Big) \ge h^2\Big[\tfrac{2}{h}\textstyle\sum_i g(s_i) - I_\rho(\sigma) + \tfrac{3}{2} - \eta_1(h)\Big],$$

*with $\eta_1(h) = \frac{|\log\rho|}{h} + \frac{2\log(1+8h/3)}{h} + \varepsilon_2(h) + \frac{\log^+ C_E}{h} \to 0$, where $C_E = \sup_{\Gamma_h}|E|e^{2\pi|\operatorname{Im} z|} < \infty$ is independent of $h$ and $\varepsilon_2(h) = O((\log h)/h)$.*

*Proof (this is the case $S = \{0, \dots, h-1\}$ of the proof of Theorem 7.1).* The exponent is $2h\log h + h^2 H$, and the term $2h\log h \ge 0$ is dropped. We write $H = H_1 + H_2 + H_3 + H_4$. ($\mathrm{H}_1$) The $h$ points $s_i$ are distinct, so by

Lemma 4.3 and Corollary 4.7, $\frac{1}{h^2}\sum_{i\neq j}-\log|s_i-s_j| \geq \frac{1}{h^2}\sum_{i\neq j}k_\rho(s_i,s_j) \geq \frac{2}{h}\sum_i U^\sigma_\rho(s_i) - I_\rho(\sigma) - \frac{|\log\rho|}{h}$. ($\mathrm{H}_2$) Since $\operatorname{Re} z_i \leq -3/8$, $t\mapsto\log|s-t|$ is increasing on $[0,1]$ and the left Riemann sum for $U^\tau$ loses at most $\log(1+1/|s|)/h \leq \log(1+8h/3)/h$; this gives $\frac{2}{h}\sum_i(-U^\tau(s_i)) - \frac{2\log(1+8h/3)}{h}$. ($\mathrm{H}_3$) With $t_a = a/h$, $\frac{1}{h^2}\sum_{a\neq b}-\log|t_a-t_b| = \frac{1}{h^2}[h(h-1)\log h - 2\sum_{n<h}\log n!] \geq \frac{3}{2} - \varepsilon_2(h)$, using $\log n! \leq (n+\frac{1}{2})\log n - n + 1$. ($\mathrm{H}_4$) $-\log|E|^{1-\kappa} \geq (1-\kappa)(2\pi|\operatorname{Im} z| - \log^+ C_E)$, with $C_E$ independent of $h$ because $E$ is 1-periodic and $x^* \equiv 5/8 \pmod 1$. Since $|\operatorname{Im} z| = h|\operatorname{Im} s|$, the sum of the field terms is $h^2\cdot\frac{2}{h}\sum_i(1-\kappa)\pi|\operatorname{Im} s_i| - h\log^+ C_E$. Adding ($\mathrm{H}_1$)–($\mathrm{H}_4$) gives the claim. □

**Lemma 4.9** (finite-cell bound $B_\rho$)**.** *Let $h \geq 2^{30}$ and let $s_1,\dots,s_k$ be the scaled points of any admissible configuration of $k \leq h$ points of $\Gamma_h \cup (-\mathcal{A})$ (atoms pairwise distinct). Then*

$$\frac{1}{h}\sum_{i\leq k} g(s_i) \;\geq\; \frac{k}{h}\,m' + S_{\mathrm{cert}} - \frac{2}{h}\sum_\ell (m'-\gamma_\ell)^+.$$

*Proof.* Write $\frac{1}{h}\sum_i g(s_i) = \frac{k}{h}m' + \frac{1}{h}\sum_i(g(s_i)-m')$. *Contour points.* Write $x^*/h = -r_0 + \delta_h$, $\delta_h = (r_0 h - N_0 - \frac{3}{8})/h \in [-\frac{3}{8h}, \frac{5}{8h})$, so $|\delta_h| < 1/h \leq \omega$ and $s_i = -r_0 + \delta_h + u(-1 \pm 3i)$. For $u \leq u_{\max}$ the point $s_i$ (or its conjugate; $g$ is conjugation-symmetric) lies in a box of (R), and for $u > u_{\max}$ the tail bound applies; in either case $g(s_i) \geq m'$. *Atoms.* Atoms lie in $-(\mathbb{Z}_{\geq 0} + \frac{3}{4})/h \cup -\mathbb{Z}_{\geq 1}/h \subset [-r_0, 0]$ and are distinct. Assign each to one cell containing it (an atom on a shared endpoint is assigned once). The two progressions have spacing $1/h$, so a closed cell $C_\ell$ receives $N_\ell \leq 2h(b_\ell - a_\ell)+2$ atoms. Since $g \geq \gamma_\ell$ on $C_\ell$ and $\min(\gamma_\ell - m', 0) \leq 0$, $\frac{1}{h}\sum_{\mathrm{atoms}}(g-m') \geq \sum_\ell \frac{N_\ell}{h}\min(\gamma_\ell - m', 0) \geq S_{\mathrm{cert}} - \frac{2}{h}\sum_\ell(m'-\gamma_\ell)^+$. Here $N_\ell/h \leq 2(b_\ell - a_\ell)+2/h$ and $2\sum_\ell(b_\ell - a_\ell)\min(\gamma_\ell - m', 0) = S_{\mathrm{cert}}$. □

**Theorem 4.10** (decay; computer-assisted)**.** $\det\mathcal{G}_h \leq \exp(-(D^+ - o(1))h^2)$ *with $D^+ = D_{\mathrm{cert}} \geq 3.1113790$. Hence $|P_h(\lambda)| \leq \exp(-(D^+ - o(1))h^2)$.*

*Proof.* Split $d|m| = w_\kappa\, d\Omega_\kappa$ with $\Omega_\kappa = \sum_{\mathcal{A}}\delta + \frac{1}{2\pi}|E|^\kappa|dz|$, of total mass $\leq 2h + c_\kappa$, $c_\kappa = \frac{1}{2\pi}\int_{\Gamma_0}|E|^\kappa|dz| < \infty$ independent of $h$ by periodicity. By Lemmas 4.8 and 4.9 (with $k = h$), the integrand of Proposition 4.2 against $\prod w_\kappa$ is at most $\exp(-h^2[D_{\mathrm{cert}} - \eta_1 - 2\eta_2])$ pointwise, where $\eta_2(h) = \frac{2}{h}\sum_\ell(m'-\gamma_\ell)^+$ (the cell number is fixed, so this is $O(1/h)$). Therefore

$$\det\mathcal{G}_h \leq \frac{1}{h!}(2h + c_\kappa)^h \exp(-h^2[D_{\mathrm{cert}} - \eta_1 - 2\eta_2]),$$

and $\log(2h + c_\kappa)/h \to 0$. Finally $|P_h(\lambda)| \leq 16^{-h}\det\mathcal{G}_h$ by Proposition 4.2. □

*Remark* 4.11. The error terms $\eta_1, \eta_2$ are $O((\log h)/h)$ with moderate explicit constants; no explicit $h_0$ is claimed (the certificate itself requires $h \geq 2^{30}$). The exponent $D^+ \geq 3.1113790$ is certified; numerics suggest that the true

limit is $\frac{9}{2}\log 2 = 3.11916\ldots$ (not used). The margin for the deductions below is

$$\eta := D^+ - \Lambda \;\geq\; 3.1113790 - 2.9809524 = 0.1304266,$$

and the argument of §8 uses the decay theorem only through this margin.

*Remark* 4.12 (cross-check by circle smoothing). An earlier version of the energy step replaced each point by a uniform measure on a circle of radius $1/h$ and used a certificate measure $\sigma$ supported on 448 disjoint circles together with the inequality $I(\nu-\sigma) \geq 0$ for signed measures of mass zero [12, Lemma I.1.8] and a Lipschitz bound for $U^\sigma$ ($L_\sigma \leq 1522.04$). That route, with an independent interval-arithmetic certificate, gives $D^+ \geq 3.1072419$, $m' \geq 1.9681060$, $S \geq -0.1101022$, $g_{\min} \geq -1.4643531$, $c \leq 6.8650$ and $\mu(\lambda) \leq 79.35$ (the value 79.3439). We keep it only as an independent cross-check of the decay exponent: the two certificates use different kernels, different measures $\sigma$ and a different ray angle ($3\pi/5$), and agree on $D^+ \approx 3.11$. The $k_\rho$ route gives the larger $D^+$ because its certificate measure was re-optimised.

## 5. $\ell$-adic nonvanishing

Let $\ell \equiv 3 \pmod 4$ be a prime with $\ell \geq 7$, $K = (\ell+1)/4$ (an integer $\geq 2$), and let $K+1 \leq h \leq \ell-1$. Let $\mathbb{Z}_{(\ell)}$ be the localization of $\mathbb{Z}$ at $\ell$. Call the indices $a < K$ *low* and $a \geq K$ *high*.

**Lemma 5.1** ($\ell$-singular structure)**.** *Among the quantities $k$, $4k-1$, $b-a$, $4$ occurring in the entries of $M_0$ ($0 \leq a < b < h$, $1 \leq k \leq h-1$) only $4k-1$ can be divisible by $\ell$, and this occurs only for $k = K$ (with $4K-1 = \ell$). Consequently every high diagonal entry of $M_0$ contains the term $+\ell^{-2}$, every low–high entry contains $-1/(16K\ell(b-a))$, and all other entries lie in $\mathbb{Z}_{(\ell)}$.*

*Proof.* $k, b-a < \ell$. If $\ell \mid 4k-1$ with $4k-1 \leq 4h-5 < 4\ell$ then $4k-1 \in \{\ell, 2\ell, 3\ell\}$; $2\ell$ is even and $3\ell \equiv 1 \pmod 4$, so $4k-1 = \ell$, i.e. $k = K$. The sum $\sum_{a<k\leq b}$ contains $k = K$ iff $a < K \leq b$ (low–high); the diagonal sum $\sum_{k\leq a}$ contains it iff $a \geq K$. □

**Lemma 5.2.** *Let $D = \operatorname{diag}(I_K, \ell I_{h-K})$. Then $D(XI+M_0)D \in M_h(\mathbb{Z}_{(\ell)}[X])$ and, modulo $\ell$,*

$$D(XI+M_0)D \equiv \begin{pmatrix} XI_K + A & \hat{B} \\ \hat{B}^\top & I \end{pmatrix}, \qquad A = M_0|_{\text{low}} \bmod \ell, \quad \hat{B}_{ab} \equiv -\frac{1}{4(b-a)} \;(a < K \leq b),$$

*and hence $\ell^{2(h-K)} P_h(X) \equiv \det(XI_K + A - \hat{B}\hat{B}^\top) \pmod \ell$. The right-hand side is monic of degree $K$; in particular $v_\ell(\operatorname{den} P_h) = 2(h-K)$.*

*Proof.* By Lemma 5.1: the high diagonal entries times $\ell^2$ are $\equiv 1$ (the term $\ell^{-2}$), the low–high entries times $\ell$ are $\equiv -1/(16K(b-a)) \equiv -1/(4(b-a))$ since $16K = 4\ell+4$, the high–high off-diagonal entries times $\ell^2$ and the term $X\ell^2$ are $\equiv 0$, and the low block is unchanged. Take the Schur complement of the identity block; $\det D = \ell^{h-K}$. □

**Lemma 5.3** (key identity)**.** *Let $s=\ell-h\in[1,\ell-K-1]$, $m=\ell-K$, and $C_s=\big[1/(a+j)\big]_{0\le a<K,\ 1\le j\le s}$ (entries in $\mathbb{F}_\ell$; note $a+j\le K-1+s<\ell$). Put $c=-H_m^{(2)}\in\mathbb{F}_\ell$ and $Y=16X+c$. Then, in $\mathbb{F}_\ell[X]$,*

$$16^K\ell^{2(h-K)}P_h(X)\equiv\det(Y\,I_K+C_sC_s^{\top}).$$

*Proof.* Modulo $\ell$, $4K=\ell+1\equiv1$, so $\frac{4}{4k-1}=\frac{1}{k-1/4}\equiv\frac{1}{k-K}\equiv\frac{1}{k+m}$ and thus $-\frac{1}{k(4k-1)}\equiv\frac1k-\frac{1}{k+m}$ and $\frac{16}{(4k-1)^2}\equiv\frac{1}{(k+m)^2}$. For $a<b<K$ (so $b+m<\ell$) summing over $a<k\le b$ and using $\frac{1}{(b-a)}\big(\frac{1}{a+j}-\frac{1}{b+j}\big)=\frac{1}{(a+j)(b+j)}$,

$$16A_{ab}\equiv\frac{1}{b-a}\sum_{k=a+1}^{b}\Big(\frac1k-\frac{1}{k+m}\Big)=\sum_{j=1}^{m}\frac{1}{(a+j)(b+j)},\qquad 16A_{aa}\equiv\sum_{k\le a}\Big(\frac{1}{(k+m)^2}-\frac{1}{k^2}\Big)=\sum_{j=1}^{m}\frac{1}{(a+j)^2}$$

Thus $16A\equiv cI+C_mC_m^{\top}$. Next, $16(\hat B\hat B^{\top})_{aa'}\equiv\sum_{b=K}^{h-1}\frac{1}{(b-a)(b-a')}$. Substituting $j=\ell-b$ (which runs over $s+1,\dots,m$ as $b$ runs over $h-1,\dots,K$) we have $a+j\equiv-(b-a)$, so this equals $\sum_{j=s+1}^{m}\frac{1}{(a+j)(a'+j)}$. Subtracting, $16(A-\hat B\hat B^{\top})\equiv cI+C_sC_s^{\top}$, and Lemma 5.2 gives the claim, since $\det(XI+(cI+C_sC_s^{\top})/16)=16^{-K}\det(YI+C_sC_s^{\top})$. □

(This was checked by exact characteristic-polynomial reduction for $h=12,20$ and every $\ell\equiv3\pmod4$ in $(h,4h]$.)

Write $\Xi_s=C_s^{\top}C_s$, i.e. $(\Xi_s)_{ij}=\sum_{a=0}^{K-1}\frac{1}{(a+i)(a+j)}$. By Sylvester's determinant identity, for $s\le K$,

$$\det(YI_K+C_sC_s^{\top})=Y^{K-s}\det(YI_s+\Xi_s).\tag{5.1}$$

**Theorem 5.4** ($\ell$-adic nonvanishing)**.** *Let $\ell\equiv3\pmod4$ be prime, $\ell\ge7$, $K=(\ell+1)/4$, and let $r\in\mathbb{Q}$ with $\ell\nmid\operatorname{den}r$. Then at least one of*

$$h_1=\ell-1,\qquad h_2=\ell-2,\qquad h_3=\ell-K=\tfrac{3\ell-1}{4}$$

*satisfies $v_\ell\big(P_h(r)\big)=-2(h-K)$; in particular $P_h(r)\neq0$.*

*Proof.* All three $h_i$ satisfy $K+1\le h_i\le\ell-1$ (for $h_3$: $\frac{3\ell-1}{4}\ge\frac{\ell+5}{4}\iff\ell\ge3$), and have $s=1,2,K$ respectively, all $\le K$ since $K\ge2$. By Lemmas 5.2 and 5.3, $\ell^{2(h-K)}P_h(r)$ is an $\ell$-adic unit iff the polynomial (5.1) does not vanish at $Y=y:=16r+c\in\mathbb{F}_\ell$. Suppose it vanishes at $y$ for all three values of $s$.

- $s=K$: at $Y=0$ the value of (5.1) is $\det(\Xi_K)=\det(C_K)^2$, where $C_K=[1/(a+j)]$ is a Cauchy matrix with $\det C_K=\prod_{a<a'}(a'-a)\prod_{j<j'}(j'-j)/\prod_{a,j}(a+j)$, a ratio of products of integers in $[1,2K-1]$, and $2K-1<\ell$. So $\det C_K\neq0$ in $\mathbb{F}_\ell$, the polynomial does not vanish at 0, and hence $y\neq0$.
- $s=1$: the value is $y^{K-1}(y+\Xi_{11})$ with $\Xi_{11}=\sum_{a<K}(a+1)^{-2}=H_K^{(2)}$. As $y\neq0$, $y=-\Xi_{11}\neq0$.
- $s=2$: the value is $y^{K-2}\det\begin{pmatrix}0&\Xi_{12}\\ \Xi_{12}&\Xi_{22}-\Xi_{11}\end{pmatrix}=-y^{K-2}\Xi_{12}^2$, and $\Xi_{12}=\sum_{a<K}\big(\frac{1}{a+1}-\frac{1}{a+2}\big)=\frac{K}{K+1}\neq0$ in $\mathbb{F}_\ell$ since $1\le K<K+1<\ell$. This is nonzero, a contradiction.

(If $\ell = 7$ then $h_2 = h_3$, $K = 2$, and the argument is unchanged.) Hence for some $i$ the reduction is nonzero, so $v_\ell(P_{h_i}(r)) = -2(h_i - K)$. □

(Exact characteristic polynomials for $\ell = 7, 11, 19, 23, 31, 43$ confirm that the three reductions have gcd = 1 and the stated valuations.)

**Lemma 5.5** (primes dividing the denominator)**.** *Let $\ell \equiv 3 \pmod 4$ be prime, $\ell \geq 7$, $K + 1 \leq h \leq \ell - 1$, and $r = p/q$ with $e = v_\ell(q) \geq 1$; put $u = \ell^e r \in \mathbb{Z}_{(\ell)}^\times$. If $e = 1$, or $e \geq 3$, or $e = 2$ and $u \not\equiv -1 \pmod \ell$, then $P_h(r) \neq 0$ for every admissible $h$.*

*Proof.* Let $N = D(rI + M_0)D$. By Lemmas 5.1, 5.2, its blocks are $rI + M_{0,LL}$, $\ell M_{0,LH} \equiv \hat{B}$, and $\ell^2 rI + J$ with $J \equiv I$, all entries other than the $r$'s being $\ell$-integral. Multiply the low rows by $\ell^e$ (and, if $e \geq 3$, the high rows by $\ell^{e-2}$). Modulo $\ell$ the matrix becomes $\begin{pmatrix} uI & 0 \\ \hat{B}^\top & I \end{pmatrix}$ for $e = 1$ (determinant $u^K$), $\begin{pmatrix} uI & 0 \\ \hat{B}^\top & (u+1)I \end{pmatrix}$ for $e = 2$ (determinant $u^K(u+1)^{h-K}$), $uI_h$ for $e \geq 3$ (determinant $u^h$). These are units in the stated cases. □

(Checked on 1044 exact cases; the exception $e = 2$, $u \equiv -1$ does occur.)

5.1. **Proof of Theorem 1.1.** Suppose $\lambda = a/b$ in lowest terms. Choose a prime $\ell \equiv 3 \pmod 4$, $\ell \geq 7$, $\ell \nmid b$, as large as we please (there are infinitely many primes $\equiv 3 \pmod 4$). Theorem 5.4 gives $h \geq (3\ell - 1)/4$ with $P_h(\lambda) \neq 0$. By Lemma 2.4, Theorem 3.1 and Theorem 4.10,

$$1 \leq b^h \operatorname{den}(P_h)\, |P_h(\lambda)| \leq \exp\bigl(h \log b - (D^+ - \Lambda - o(1))h^2\bigr),$$

which is false for large $h$ because $D^+ - \Lambda > 0.130$. □

## 6. A determinant evaluation and the choice of $h$

Theorem 5.4 uses three values of $h$ whose ratio $h_{\max}/h_{\min}$ is $4/3$ (the factor $4/3$ in the exponent of Theorem 8.1). In this section we show that already the two values

$$h \in \{\ell - K,\ \ell - K - 1\} \qquad (\text{that is, } s = \ell - h \in \{K, K+1\})$$

suffice; their ratio is $1 + O(1/\ell)$. The proof rests on a statement about two consecutive Gram-type matrices over $\mathbb{F}_\ell$ (Lemma CC below), which in turn reduces to the non-vanishing modulo $\ell$ of an explicit Krylov determinant; the latter is evaluated in closed form (Theorem A). By (5.1) the polynomial $\det(YI_K + C_sC_s^\top)$ has the factor $Y^{K-s}$ for $s < K$, so the value $y = 0$ cannot be excluded for any $h > \ell - K$; this is why the smallest usable $s$ is $K$.

Throughout this section $\ell \equiv 3 \pmod 4$ is a prime, $\ell \geq 7$, $K = (\ell + 1)/4$, and for $s \geq 0$ with $K + s < \ell$ we use, over $\mathbb{F}_\ell$,

$$C_s = \Bigl[\frac{1}{a+j}\Bigr]_{0 \leq a < K,\ 1 \leq j \leq s}, \qquad N_s = C_sC_s^\top \in M_K(\mathbb{F}_\ell), \qquad w_j = \Bigl(\frac{1}{a+j}\Bigr)_{0 \leq a < K},$$

so that $N_{s+1} = N_s + w_{s+1}w_{s+1}^\top$. All denominators are $\ell$-units because $a+j \le K-1+s+1 < \ell$.

### 6.1. **Statement.**

**Lemma 6.1** (Lemma CC)**.** *Let $s \ge K$ with $K+s<\ell$. Then $N_s$ and $N_{s+1}$ have no common eigenvalue in $\overline{\mathbb{F}}_\ell$. Equivalently, the polynomials $\det(YI_K + N_s)$ and $\det(YI_K + N_{s+1})$ have no common root in $\overline{\mathbb{F}}_\ell$. In particular this holds for $s = K$ (note $2K < \ell$).*

The proof (§6.6) is the combination of the Krylov reduction of §6.2 with the evaluation of Theorem A below.

**Theorem 6.2** (Choice of $h$)**.** *Let $\ell \equiv 3 \pmod 4$ be prime, $\ell \ge 7$, $K = (\ell+1)/4$, and let $r \in \mathbb{Q}$ with $\ell \nmid \operatorname{den} r$. Then at least one of*

$$h_1 = \ell - K = \tfrac{3\ell-1}{4}, \qquad h_2 = \ell - K - 1 = \tfrac{3\ell-5}{4}$$

*satisfies $v_\ell\big(P_h(r)\big) = -2(h-K)$; in particular $P_h(r) \neq 0$.*

### 6.2. **The Krylov reduction.**

**Lemma 6.3.** *Let $F$ be a field of characteristic $\neq 2$, $N \in M_K(F)$ symmetric, $w \in F^K$. If $z \in F$ is a common root of $\det(z-N)$ and $\det(z-N-ww^\top)$, then there is $x \neq 0$ with $Nx = zx$ and $w^\top x = 0$.*

*Proof.* The matrix determinant lemma gives $\det(z-N-ww^\top) = \det(z-N) - w^\top \operatorname{adj}(z-N)\, w$, so $w^\top \operatorname{adj}(z-N)\, w = 0$. Since $\det(z-N) = 0$, $\operatorname{rank}(z-N) \le K-1$. If $\operatorname{rank}(z-N) \le K-2$, then $\ker(z-N)$ has dimension $\ge 2$ and so contains some $x \neq 0$ with $w^\top x = 0$. If $\operatorname{rank}(z-N) = K-1$, then $\operatorname{adj}(z-N)$ is symmetric of rank one with columns in $\ker(z-N) = \langle x \rangle$, so $\operatorname{adj}(z-N) = \gamma\, xx^\top$ with $\gamma \neq 0$, and $\gamma(w^\top x)^2 = 0$ forces $w^\top x = 0$. □

(This covers isotropic eigenvectors, $x^\top x = 0$, which genuinely occur modulo $\ell$.)

**Lemma 6.4.** *Let $N \in M_K(F)$ be symmetric and $w \in F^K$. If some $x \neq 0$ satisfies $Nx = zx$ and $w^\top x = 0$, then $\det[w, Nw, \dots, N^{K-1}w] = 0$.*

*Proof.* $x^\top N^k w = (N^k x)^\top w = z^k x^\top w = 0$ for every $k$, so $x$ lies in the left kernel of the Krylov matrix. □

**Corollary 6.5.** *Let $s \ge 1$, $K+s<\ell$, and $u(K,s) := \det[w_{s+1}, N_s w_{s+1}, \dots, N_s^{K-1} w_{s+1}] \in \mathbb{F}_\ell$. If $u(K,s) \neq 0$, then $N_s$ and $N_{s+1}$ have no common eigenvalue in $\overline{\mathbb{F}}_\ell$.*

*Proof.* Apply Lemmas 6.3 and 6.4 over $F = \overline{\mathbb{F}}_\ell$ with $N = N_s$ and $w = w_{s+1}$. □

6.3. **Theorem A.** For real $L, \rho \notin \{0, -1, -2, \dots\}$ and $n \geq 1$ let $c_y = (1/(a+y))_{a=0}^{n-1} \in \mathbb{R}^n$ and

$$N(L,\rho) := \sum_{k\geq 0} \left[ c_{L+k} c_{L+k}^{\top} - c_{\rho+k} c_{\rho+k}^{\top} \right] \in M_n(\mathbb{R}),$$

which converges absolutely (each term is $O(k^{-3})$). We write $\theta = L - 1$, $s = \rho - L$ (so $\rho = \theta + s + 1$), and

$$u_n(\theta, s) := \det\left[w, Nw, \dots, N^{n-1}w\right], \qquad N = N(\theta{+}1, \theta{+}s{+}1),\ w = c_{\theta+s+1}.$$

**Lemma 6.6.** *(a) $N(L,\rho)$ is real symmetric, meromorphic in $(L,\rho)$, with poles only where $L+k+a=0$ or $\rho+k+a=0$ for some $0 \leq a < n$, $k \geq 0$; near any point only finitely many terms are singular. (b) If $s = \rho - L \in \mathbb{Z}_{\geq 0}$, then $N(L,\rho) = \sum_{y=L}^{\rho-1} c_y c_y^{\top}$ (a sum of $s$ terms); if $L - \rho = c \in \mathbb{Z}_{\geq 0}$, then $N(L,\rho) = -\sum_{y=\rho}^{L-1} c_y c_y^{\top}$. (c) $N(L,\rho+1) = N(L,\rho) + c_\rho c_\rho^{\top}$ and $N(L,\rho) = c_L c_L^{\top} + N(L+1,\rho)$. (d) Off-diagonal entries are rational functions of $(L,\rho)$, and $N(L,\rho) = N^{\mathrm{rat}}(L,\rho) + (\psi'(L) - \psi'(\rho))\, I$ with $N^{\mathrm{rat}}$ rational. (e) $u_n$ is a rational function of $(\theta, s)$. (f) For integer $s \geq 0$ and $\theta = 0$: $N(1, s{+}1) = N_s$ (the matrix of §6.2, over $\mathbb{Q}$) and $w = c_{s+1} = w_{s+1}$; thus $u_n(0,s) = u(n,s)$ with $K = n$.*

*Proof.* (a)–(c) are immediate from the definition (telescoping). For (d), $\sum_{k\geq 0} \frac{1}{(a+y+k)(b+y+k)} = \frac{1}{b-a} \sum_{j=0}^{b-a-1} \frac{1}{a+y+j}$ for $a < b$, and $\sum_{k\geq 0} (a+y+k)^{-2} = \psi'(y) - \sum_{j<a} (y+j)^{-2}$, i.e. $\psi'(y)$ minus a rational function. For (e), the Krylov determinant is invariant under $N \mapsto N + \tau I$ (the change of basis from $N^k w$ to $(N+\tau)^k w$ is unitriangular), so $\psi'(L) - \psi'(\rho)$ can be dropped, and $u_n$ is a polynomial in the entries of $N^{\mathrm{rat}}$ and $w$. (f) is (b). □

**Theorem 6.7** (Theorem A)**.** *Put $T(m) = m(m+1)/2$ and*

$$A_n(s) = \prod_{c=1}^{n-1} \left[(s+c)(s+1-c)\right]^{T(n-c)}, \qquad B_n(\theta) = \prod_{i=1}^{n-1} (\theta+i)^{2i(n-i)},$$

$$C_n(t) = \prod_{c=1}^{n} (t+c)^{\beta_c}, \quad \beta_c = 2c(n+1-c) - n, \qquad \kappa_n^{+} = \prod_{m=1}^{n-1} m^{2(n-m)(n-2m)}.$$

*Then there is a constant $\kappa_n \in \mathbb{R}$ with $|\kappa_n| = \kappa_n^{+}$ such that, as rational functions of $(\theta, s)$,*

$$u_n(\theta, s) = \kappa_n \, \frac{A_n(s)}{B_n(\theta)\, C_n(\theta + s)}.$$

*Remark* 6.8. *Only $|u_n|$ is needed.* Numerically $\kappa_n = (-1)^{\lfloor n/2 \rfloor} \kappa_n^{+}$; the sign is not proved here and is not used. The formula was also verified exactly (rational arithmetic) for every $n \leq 12$, at random rational $(\theta, s)$ including non-integer and negative $s$, and with the Cauchy Gram matrices $N_s$ themselves (§10). For $n = 3$, $\theta = 0$ it gives $u(3,s) = -s^3(s-1)/(64(s+2)^4(s+3)^3)$.

6.4. **Tools.**

**Lemma 6.9** (resultant identity)**.** *Let $N$ be real symmetric $n\times n$ with eigenvalues $\lambda_1,\dots,\lambda_n$, let $w\in\mathbb{R}^n$, and let $N'=N+ww^\top$ have eigenvalues $\mu_1,\dots,\mu_n$. Then*

$$\det[w,Nw,\dots,N^{n-1}w]^2=\pm\prod_{i,k}(\lambda_i-\mu_k).$$

*Proof.* Write $N=U\,\mathrm{diag}(\lambda)U^\top$ with $U$ orthogonal and $\tilde w=U^\top w$. Then $\det[w,Nw,\dots]=\det U\cdot\prod_k\tilde w_k\cdot\prod_{i<j}(\lambda_j-\lambda_i)$. On the other hand $\chi_{N'}(z)=\chi_N(z)(1-\sum_k\tilde w_k^2/(z-\lambda_k))$, so $\chi_{N'}(\lambda_k)=-\tilde w_k^2\prod_{j\neq k}(\lambda_k-\lambda_j)$, and $\prod_{i,k}(\lambda_i-\mu_k)=\pm\prod_k\chi_{N'}(\lambda_k)=\pm\prod_k\tilde w_k^2\prod_{i<j}(\lambda_i-\lambda_j)^2$. (This is the classical identity for the resultant of the characteristic polynomials of $N$ and $N'$.) □

We call this identity (R). It exhibits $|u_n|^2$ as a product over pairs $(i,k)$ of the differences $|\lambda_i-\mu_k|$.

**Lemma 6.10** (cluster lemma)**.** *Let $M(\varepsilon)=\varepsilon^{-2}\sigma J+\varepsilon^{-1}Y+Z(\varepsilon)$ be real symmetric $n\times n$ matrices, $\varepsilon\to0$, where $\sigma=\pm1$, $J=I_r\oplus0_{n-r}$, $Y$ is real symmetric with both diagonal blocks (the $r\times r$ and the $(n-r)\times(n-r)$ block) equal to zero, and $Z(\varepsilon)$ is bounded. Then $M(\varepsilon)$ has $r$ eigenvalues $\sigma\varepsilon^{-2}+O(1)$ and $n-r$ eigenvalues $O(1)$.*

*Proof.* Write $\varepsilon^2M=\sigma J+\varepsilon Y+O(\varepsilon^2)$. Let $S=\begin{pmatrix}0&X\\-X^\top&0\end{pmatrix}$ with $X=-\sigma Y_{12}$ ($Y_{12}$ the off-diagonal block of $Y$), and $Q=\exp(\varepsilon S)$, which is orthogonal. Since $[J,S]=\begin{pmatrix}0&X\\X^\top&0\end{pmatrix}$,

$$Q^\top(\varepsilon^2M)Q=\sigma J+\varepsilon\big(Y+\sigma[J,S]\big)+O(\varepsilon^2)=\sigma J+O(\varepsilon^2),$$

because the diagonal blocks of $Y$ vanish and the off-diagonal block of $Y+\sigma[J,S]$ is $Y_{12}+\sigma X=0$. Weyl's inequality $|\lambda_i(X_1+X_2)-\lambda_i(X_1)|\le\|X_2\|$ shows that $\varepsilon^2M$ has $r$ eigenvalues $\sigma+O(\varepsilon^2)$ and $n-r$ eigenvalues $O(\varepsilon^2)$. □

6.5. **Proof of Theorem A.** Throughout, $\theta\in\mathbb{R}\setminus\mathbb{Z}$ is fixed (generic) unless said otherwise, so that the points $s=-\theta-c$ (poles) and the integers $s=j$ (zeros) below are distinct. We write $\lambda_i$, $\mu_k$ for the eigenvalues of $N$ and $N'=N+ww^\top$ and use (R) throughout.

**Step 1: zeros in $s$.** *At $s=j$, $0\le j\le n-2$.* By Lemma 6.6(b), $N$ is a sum of $j$ rank-one terms, so $0$ is an eigenvalue of $N$ of multiplicity $\ge n-j$ and of $N'$ of multiplicity $\ge n-j-1$. Near $s=j$ the matrix $N$ is analytic in $s$ (for $\theta\notin\mathbb{Z}$ no term is singular), so by Weyl's inequality these eigenvalues are $O(|s-j|)$. This gives at least $(n-j)(n-j-1)$ factors of size $O(|s-j|)$ in (R), all other factors being bounded; hence $u_n$ vanishes at $s=j$ to order $\ge T(n-1-j)$. *At $s=-c$, $1\le c\le n-1$.* By Lemma 6.6(b), $N=-\sum_{y=\rho}^{L-1}c_yc_y^\top$ has rank $\le c$, and $w=c_\rho$ is one of these terms, so $N'$ has rank $\le c-1$. The same argument gives $(n-c)(n-c+1)$ small

factors and order $\geq T(n-c)$. Since $A_n(s)$ vanishes exactly at these points to exactly these orders ($s+1-c=0$ at $s=j=c-1$, with exponent $T(n-c)=T(n-1-j)$), $u_n/A_n$ has no pole in $s$ at the zeros of $A_n$.

**Step 2: poles in $s$.** For fixed $\theta$ the entries of $N$ and $w$ are singular only at $\rho = 1-c$, i.e. $s=-\theta-c$, $c\geq 1$. Near $\rho = 1-c+\varepsilon$, the singular terms are $c_{\rho+k}c_{\rho+k}^\top$ with $k=c-1-a$ for $a$ in the set $\{0,\dots,\min(c,n)-1\}$ (call its size $c^*=\min(c,n)$); entry $a$ of $c_{\rho+k}$ has a simple pole, so each singular term contributes $-\varepsilon^{-2}e_ae_a^\top$ plus an $\varepsilon^{-1}$ term plus a bounded part. Thus

$$N = -\varepsilon^{-2}J_{c^*} + \varepsilon^{-1}Y + O(1).$$

The $\varepsilon^{-1}$-coefficient $Y$ has vanishing diagonal blocks: for $p\neq q$ in the singular set, the contributions $1/(q-p)$ (from the term singular at $p$) and $1/(p-q)$ (from the term singular at $q$) cancel; the diagonal entries have no $\varepsilon^{-1}$ term since $(\varepsilon+m)^{-2}$ is regular for $m\neq 0$; and entries with both indices outside the singular set are regular. *Case $c>n$.* Then $J_{c^*}=I$, $w$ is regular, and $N$, $N'$ are $-\varepsilon^{-2}I+\varepsilon^{-1}Y+O(1)$ with $Y=0$ (only the first block exists); all eigenvalues of both are $-\varepsilon^{-2}+O(1)$, all differences are $O(1)$, so $u_n$ is bounded: no pole. *Case $c\leq n$.* Here $w=c_\rho$ has the pole in coordinate $a=c-1$, and $ww^\top$ is exactly the term $k=0$ of the $\rho$-series, so $N'=N+ww^\top$ has the same form with $J_{c-1}$ (and the same vanishing of the diagonal blocks of its $\varepsilon^{-1}$ coefficient). By Lemma 6.10 (with $\sigma=-1$), $N$ has $c$ "big" eigenvalues $-\varepsilon^{-2}+O(1)$ and $n-c$ "small" ones $O(1)$, and $N'$ has $c-1$ big and $n-c+1$ small ones. In (R), big–big and small–small differences are $O(1)$, and each of the

$$c(n-c+1)+(c-1)(n-c) = 2c(n+1-c)-n = \beta_c$$

big–small differences has size $\varepsilon^{-2}$. Hence $u_n = O(\varepsilon^{-\beta_c})$.

**Step 3: poles in $\theta$ at $\theta=-i$, $1\leq i\leq n-1$, $s$ generic.** Put $L=1-i+\varepsilon$. The singular terms are $c_{L+k}c_{L+k}^\top$, $k=i-1-a$, $a\in\{0,\dots,i-1\}$, and exactly as in Step 2, but with sign $+$, $N=+\varepsilon^{-2}J_i+\varepsilon^{-1}Y+O(1)$ with $Y$ having vanishing diagonal blocks. Now $w=c_\rho$ is regular, so $N'$ has the same singular part. Both have $i$ big and $n-i$ small eigenvalues, giving $2i(n-i)$ big–small differences, so $u_n=O(\varepsilon^{-2i(n-i)})$.

**Step 4: identification in $s$.** As $\rho\to\infty$ with $L$ fixed, $N=R_\infty(L)+O(1/\rho)$ with $R_\infty(L)=\sum_{k\geq 0}c_{L+k}c_{L+k}^\top$, analytic in $1/\rho$, and $\rho w=\mathbf{1}+O(1/\rho)$ with $\mathbf{1}$ the all-ones vector. Hence $\rho^n u_n\to\det[\mathbf{1},R_\infty\mathbf{1},\dots,R_\infty^{n-1}\mathbf{1}]$, finite, and $u_n=O(s^{-n})$. Consider $F(s):=u_nC_n(\theta+s)/A_n(s)$ as a rational function of $s$. By Steps 1–2, $F$ has no poles (the poles of $u_n$ are at $s=-\theta-c$, $c\leq n$, of order $\leq\beta_c$, where $A_n\neq 0$; the zeros of $A_n$ are integers, where $u_n$ vanishes to at least the order of $A_n$ and $C_n(\theta+s)\neq 0,\infty$), so $F$ is a polynomial in $s$. Both $A_n$ and $C_n$ are monic and

$$\deg C_n - \deg A_n = \left[\tfrac{n(n+1)(n+2)}{3}-n^2\right]-\tfrac{(n-1)n(n+1)}{3} = n,$$

so $F(s) = O(s^{-n}s^n) = O(1)$, and $F$ is a constant $K_n(\theta)$:

$$u_n(\theta, s) = K_n(\theta)\,\frac{A_n(s)}{C_n(\theta+s)}. \tag{6.1}$$

Moreover $K_n(\theta) = \lim_{s\to\infty} s^n u_n = \det[\mathbf{1}, R_\infty\mathbf{1}, \dots]$ with $L = \theta + 1$.

**Step 5: identification in $\theta$.** By Lemma 6.6(d), $K_n$ is a rational function of $\theta$ (the scalar $\psi'(L)$ in $R_\infty$ drops out of the Krylov determinant), and its entries $\sum_{k\ge0} c_{L+k}c_{L+k}^\top - \psi'(L)I$ have poles only where $L + j = 0$ for some $0 \le j \le n-2$, that is $\theta = -i$, $1 \le i \le n-1$. Comparing (6.1) with Step 3 (at $\theta = -i$ and generic $s$, $A_n(s)/C_n(\theta+s)$ is finite and nonzero), the pole of $K_n$ at $-i$ has order $\le 2i(n-i)$. Hence $K_nB_n$ is a polynomial in $\theta$. Now let $L \to \infty$ with $\rho$ fixed and generic. Then $N(L,\rho) \to -\sum_{k\ge0} c_{\rho+k}c_{\rho+k}^\top$ (the $L$-series is $O(1/L)$) and $w = c_\rho$ is fixed, so $u_n$ stays bounded. In (6.1), $s = \rho - L$ and $C_n(\theta+s) = C_n(\rho-1)$ is a fixed nonzero number, while $A_n(\rho - L)$ is of exact order $L^{\deg A_n}$. Hence $K_n(\theta) = O(\theta^{-\deg A_n})$. Since $\deg B_n = \sum_i 2i(n-i) = \frac{(n-1)n(n+1)}{3} = \deg A_n$, the polynomial $K_nB_n$ is bounded, hence a constant $\kappa_n$. This proves the shape of Theorem A:

$$u_n = \kappa_n \frac{A_n(s)}{B_n(\theta)\,C_n(\theta+s)}. \tag{6.2}$$

**Step 6: the constant, by recursion in $n$.**

**Lemma 6.11** (recursion)**.** *Fix non-integer $s$. As $\theta = -1+\varepsilon \to -1$,*

$$\varepsilon^{2(n-1)}\,|u_n(-1+\varepsilon, s)| \;\longrightarrow\; \frac{1}{|s|}\,|u_{n-1}(1,\, s-1)| \qquad (n \ge 2),$$

*where on the right $u_{n-1}$ is the $(n-1)$-dimensional function evaluated at $\theta' = 1$, $s' = s-1$.*

*Proof.* Here $L = \varepsilon$ and $\rho = s + \varepsilon$. We have $c_L = \varepsilon^{-1}p$ with $p = e_0 + O(\varepsilon)$, and by Lemma 6.6(c) $N = \varepsilon^{-2}pp^\top + A(\varepsilon)$ with $A(\varepsilon) = N(1+\varepsilon, \rho)$ analytic at $\varepsilon = 0$. Put $N' = N + ww^\top = \varepsilon^{-2}pp^\top + A'(\varepsilon)$, $A' = A + ww^\top$; $w = c_\rho$ has first coordinate $w_0 = 1/\rho \to 1/s$. *Big eigenvalues.* The largest eigenvalues are $\lambda_{\text{big}} = \varepsilon^{-2}|p|^2 + \hat p^\top A\hat p + O(\varepsilon^2)$ and $\mu_{\text{big}} = \varepsilon^{-2}|p|^2 + \hat p^\top A'\hat p + O(\varepsilon^2)$ with $\hat p = p/|p| \to e_0$; so $\mu_{\text{big}} - \lambda_{\text{big}} \to w_0^2 = 1/s^2$. *Small eigenvalues.* $\varepsilon^2\chi_N(z) = \varepsilon^2\det(z-A) - p^\top \operatorname{adj}(z-A)p \to -\det(z - A(0)[1{:}])$, where $A(0)[1{:}]$ is the compression of $A(0) = N(1,s)$ to the coordinates $1,\dots,n-1$; likewise for $N'$ with $A'(0)[1{:}] = A(0)[1{:}] + \tilde w\tilde w^\top$, $\tilde w = (w_a)_{a\ge1}$. Hence the other $n-1$ eigenvalues of $N$ (resp. $N'$) converge to those of $A(0)[1{:}]$ (resp. $A'(0)[1{:}]$). Re-indexing $a' = a-1$, the entries of $A(0)[1{:}]$ are $\sum_k\big[\frac{1}{(a'+2+k)(b'+2+k)} - \frac{1}{(a'+s+1+k)(b'+s+1+k)}\big] = N(2,s+1)_{a'b'}$ and $\tilde w = c_{s+1}$, that is, the $(n-1)$-dimensional problem with $L' = 2$, $\rho' = s+1$, i.e. $\theta' = 1$, $s' = s-1$. Now apply (R). The big–big difference tends to $1/s^2$; each of the $2(n-1)$ big–small differences is $\pm\varepsilon^{-2}(1+o(1))$; the small–small differences tend to

the corresponding product for the $(n-1)$-dimensional problem, which is $\pm u_{n-1}(1, s-1)^2$ by (R). Thus $|u_n|^2 \sim s^{-2}\varepsilon^{-4(n-1)} u_{n-1}(1, s-1)^2$. □

**Lemma 6.12** (exponent bookkeeping)**.** *$C_n(s-1)\,A_{n-1}(s-1) = s\,C_{n-1}(s)\,A_n(s)$ as polynomials in $s$.*

*Proof.* Compare the exponent of $(s+d)$ on both sides ($T(m) = 0$ for $m \le 0$): for $1 \le d \le n-3$: $2(d+1)(n-d)-n+T(n-2-d) = 2d(n-d)-n+1+T(n-d)$; $d = n-2$: both sides equal $3n-4$; $d = n-1$: both equal $n$; $d = 0$: $n + T(n-2) = 1 + T(n-1)$; $d < 0$ (the factors $s+1-c$): both equal $T(n-1+d)$. □

(The identity of Lemma 6.12 was also checked by exact computation for $2 \le n \le 8$.)

*Proof of Theorem A, conclusion.* Evaluate (6.2) at $\theta = -1+\varepsilon$. Since $B_n(-1+\varepsilon) = \varepsilon^{2(n-1)} \prod_{i=2}^{n-1} (i-1+\varepsilon)^{2i(n-i)}$,

$$\varepsilon^{2(n-1)}|u_n| \to \frac{|\kappa_n|\,|A_n(s)|}{\prod_{i=2}^{n-1}(i-1)^{2i(n-i)}\,|C_n(s-1)|}.$$

By Lemma 6.11 and (6.2) for $n-1$ at $(\theta', s') = (1, s-1)$, the limit is also $\dfrac{|\kappa_{n-1}|\,|A_{n-1}(s-1)|}{|s|\,B_{n-1}(1)\,|C_{n-1}(s)|}$, with $B_{n-1}(1) = \prod_{i=1}^{n-2}(i+1)^{2i(n-1-i)}$. By Lemma 6.12, $A_{n-1}(s-1)/(s\,C_{n-1}(s)) = A_n(s)/C_n(s-1)$, so all $s$-dependence cancels and

$$|\kappa_n| = |\kappa_{n-1}| \cdot \frac{\prod_{i=2}^{n-1}(i-1)^{2i(n-i)}}{\prod_{i=1}^{n-2}(i+1)^{2i(n-1-i)}} = |\kappa_{n-1}| \prod_{m=1}^{n-1} m^{2(2n-1-3m)},$$

where the exponent of $m$ is $2[(m+1)(n-m-1) - (m-1)(n-m)] = 2(2n-1-3m)$. Since $u_1 = 1/(\theta+s+1)$, $\kappa_1 = 1$, and summing the exponents of $m$ over $N = m+1, \dots, n$ gives $2\sum_{N=m+1}^{n}(2N-1-3m) = 2(n-m)(n-2m)$. Hence $|\kappa_n| = \prod_{m=1}^{n-1} m^{2(n-m)(n-2m)} = \kappa_n^+$. □ □

## 6.6. **Proofs of Lemma CC and Theorem 6.2.**

*Proof of Lemma CC.* Let $n = K$ and $s \ge K$, $K + s < \ell$. By Lemma 6.6(f) and Theorem A at $\theta = 0$,

$$u(K, s) = \kappa_K \frac{A_K(s)}{B_K(0)\,C_K(s)} \quad \text{in } \mathbb{Q},$$

with $\kappa_K$ of absolute value $\prod_{m<K} m^{2(K-m)(K-2m)}$ (exponents may be negative). The bases occurring are: the integers $m \in [1, K-1]$ (from $\kappa_K$ and $B_K(0) = \prod i^{2i(K-i)}$); $s+c$ with $1 \le c \le K$, so $s+c \in [s+1, s+K]$ (from $A_K$ and $C_K$); and $s+1-c \in [s-K+2, s] \subset [2, s]$ (from $A_K$). All are integers in $[1, K+s]$, and $K+s < \ell$, so every base is an $\ell$-unit and $u(K,s)$ is a nonzero element of $\mathbb{Z}_{(\ell)}$ that is a unit. The entries of $N_s$ and $w_{s+1}$ are $\ell$-integral, so the reduction modulo $\ell$ of $u(K,s)$ is the Krylov determinant over $\mathbb{F}_\ell$ of Corollary 6.5, which is therefore nonzero. Corollary 6.5 gives Lemma CC. □ □

Note that only $|u|$ enters: we use only that $u(K,s)$ is a nonzero rational number all of whose prime factors are $< \ell$.

*Proof of Theorem 6.2.* For $h_1 = \ell - K$ and $h_2 = \ell - K - 1$ we have $s = \ell - h = K$ and $K+1$ respectively. Both satisfy $K+1 \le h_i \le \ell - 1$ ($h_2 \ge K+1 \iff \ell \ge 5$) and $1 \le s \le \ell - K - 1$ ($K+1 \le \ell - K - 1 \iff \ell \ge 5$), so Lemmas 5.2 and 5.3 apply: $\ell^{2(h-K)} P_h(r)$ is an $\ell$-adic unit iff $D_s(Y) := \det(Y I_K + N_s)$ does not vanish at $Y = y := 16r + c \in \mathbb{F}_\ell$. Suppose $D_K(y) = D_{K+1}(y) = 0$. As in the proof of Theorem 5.4, $D_K(0) = \det(C_K)^2 \ne 0$ (Cauchy matrix with entries of size $< \ell$), so $y \ne 0$; thus $-y \ne 0$ is a common eigenvalue of $N_K$ and $N_{K+1}$, contradicting Lemma CC with $s = K$ ($2K < \ell$). Hence for some $i$ the reduction is nonzero and $v_\ell(P_{h_i}(r)) = -2(h_i - K)$. □ □

*Remark* 6.13 (numerical checks). Independently of Theorem A, the nonvanishing of $u(K,K)$ modulo $\ell$ (and hence Lemma CC for $s = K$) was verified by exact computation of the Krylov determinant for all $\ell \equiv 3 \pmod 4$ with $7 \le \ell < 4500$ (307 primes); for 39 primes $\ell < 400$ the characteristic polynomials of $N_K$ and $N_{K+1}$ over $\mathbb{F}_\ell$ were checked to be coprime, and Lemma 5.3 was checked for $\ell = 7, 11, 19, 23$ and $h = \ell - K, \ell - K - 1$ and all $X \in \mathbb{F}_\ell$. These are consistency checks, not used in the proof.

## 7. Node removal and the root-free window

7.1. **Why a small value is not enough.** Let $Q_h = \mathrm{den}(P_h) P_h$ and $|\lambda - p/q| = |t|$. The mean-value route, $|Q_h(p/q) - Q_h(\lambda)| \le |t| \max |Q_h'|$, with $\max |Q_h'|$ of size $e^{\Lambda h^2 + O(h)}$, only yields a bound of the shape $\exp(-C \log^2 q)$, which is not a finite irrationality measure. What is needed is smallness of $|P_h|$ on a whole window $|x - \lambda| \le e^{-ch}$ of *exponential radius*, together with $Q_h(p/q) \ne 0$. The second requirement is supplied by Theorem 5.4 (valid at every rational, including $p/q$). For the first, note that

$$P_h(\lambda + t) = \det(T_h + tI) = \sum_{S \subseteq \{0,\dots,h-1\}} t^{\,h - |S|} \det T_h[S], \tag{7.1}$$

by multilinearity in the columns, where $T_h[S]$ is the principal submatrix on $S$. So the window requires a bound for *every* principal minor, not only for the full determinant, and uniform in the number $j = h - |S|$ of removed nodes. The proof of Theorem 4.10 extends to principal minors; the only new feature is a cost for each removed node.

7.2. **Statement.** Retain the notation of §4.4: the certificate measure $\sigma$ with $n_\sigma = 300$ point masses, $\rho = 1/1024$, $g = U^\sigma_\rho - U^\tau + (1 - \kappa)\pi |\operatorname{Im} s|$, and the certified data $m' \ge 1.9718000062$, $S_{\mathrm{cert}} \ge -0.1107505186$, $g_{\min} \ge -1.4640996549$ and $D_{\mathrm{cert}} \ge 3.1113790$. On the real line the field term vanishes, so $g = U^\sigma_\rho - U^\tau$ there, and $g_{\min}$ bounds it below on $[0,1]$. Recall $c = 2(m' - g_{\min}) \le c_{\mathrm{cert}} := 6.8717994$.

**Theorem 7.1** (node removal $N_\rho$; computer-assisted)**.** *There are $\eta(h) = O((\log h)/h)$ and a constant $c_\kappa$, both independent of $S$, such that for every $h \geq 2^{30}$ and every $S \subseteq \{0, \dots, h-1\}$, with $k = |S|$, $j = h - k$ and $\beta = j/h$,*

$$|\det T_h[S]| \;\leq\; 16^{-k}\,(2h + c_\kappa)^k\,\exp\bigl(-h^2[D_{\mathrm{cert}} - \beta c] + h^2\eta(h)\bigr).$$

*Explicitly,* $\eta(h) = \frac{|\log\rho|}{h} + \frac{2\log(1+8h/3)}{h} + \varepsilon_2(h) + \frac{\log^+ C_E}{h} + \frac{4}{h}\sum_\ell (m' - \gamma_\ell)^+$.

For $S = \{0, \dots, h-1\}$ ($\beta = 0$) this is Theorem 4.10. The proof is that of Lemmas 4.8 and 4.9 line by line; we record what changes. (The coefficient of the last term of $\eta$ is $4/h$ because $H_S$ carries the factor $2/h$ in front of $\sum g$, so the error $\frac{2}{h}\sum_\ell (m' - \gamma_\ell)^+$ of Lemma 4.9 is doubled.)

7.3. **Proof of Theorem 7.1. Inputs.** By Proposition 4.1 the identity $L_F[a][b] = \int v_a v_b\, dm$ is entrywise, so it holds for principal submatrices: $L_F[S] = \int \phi_S \phi_S^{\top} dm$ with $\phi_S = (v_a)_{a\in S}$, and $T_h[S] = L_F[S]/16$. As $m$ has finite total variation and $|v_a| \leq 8/3$ on its support, Andréief's identity gives $\det L_F[S] = \frac{1}{k!}\int \det[v_a(z_i)]_{i\leq k, a\in S}^2 \prod dm(z_i)$ (for $k = 0$ both sides are 1), and therefore

$$|\det L_F[S]| \leq \frac{1}{k!}\int |C_S(z)|^2 \prod_{i\leq k} d|m|(z_i),$$

$$|C_S|^2 = \frac{\prod_{i\neq i'} |z_i - z_{i'}| \prod_{a\neq b\in S} |a-b|}{\prod_{i,\,a\in S} |z_i - a|^2},$$

by Cauchy's determinant. Split $d|m| = w_\kappa\, d\Omega_\kappa$ as in the proof of Theorem 4.10; then $\Omega_\kappa(\Sigma_h) \leq 2h + c_\kappa$ and $c_\kappa$ is independent of $h$. Every point of the support of $m$ has $\operatorname{Re} z \leq -\frac{3}{8}$ while the nodes lie in $[0, h-1]$, so free points and nodes never coincide. It therefore suffices to prove the pointwise bound

$$|C_S(z)|^2 \prod_i w_\kappa(z_i) \leq \exp(-h^2[D_{\mathrm{cert}} - \beta c - \eta(h)]), \tag{7.2}$$

since integrating it against $\prod \Omega_\kappa$ and using $1/k! \leq 1$ gives the theorem.

**Scaling.** Put $s_i = z_i/h$, $t_a = a/h$. Counting powers of $h$: $-k(k-1)$ from the free pairs, $-k(k-1)$ from the node pairs inside $S$ and $+2k^2$ from the cross terms, in total $2k\log h \geq 0$ in the exponent of the reciprocal, which is dropped. Hence $-\log(|C_S|^2 \prod w_\kappa) \geq h^2 H_S$ with

$$h^2 H_S = \sum_{i\neq i'} -\log|s_i - s_{i'}| + \sum_{a\neq b\in S} -\log|t_a - t_b| + 2\sum_{i,\,b\in S} \log|s_i - t_b| + \sum_i -\log w_\kappa(z_i).$$

**Completing the node set.** Let $S^c$ be the $j$ removed nodes, and let all pairs be of distinct elements. Two exact identities, from adding and removing the pairs meeting $S^c$ (for the second: the ordered pairs meeting

$S^c$ are those with first entry in $S^c$, plus those with second entry in $S^c$, minus those with both; the summand is symmetric), are

$$2\sum_{i,\,b\in S}\log|s_i-t_b| = 2\sum_{i,\,\text{all }b}\log|s_i-t_b| + 2\sum_{i,\,a\in S^c}(-\log|s_i-t_a|),$$

$$\sum_{b\neq b'\in S} = \sum_{\text{all } b\neq b'} -2\sum_{a\in S^c}\sum_{b\neq a} + \sum_{a\neq a'\in S^c} \qquad \text{(applied to } -\log|t_b-t_{b'}|).$$

Let $X$ be the set of the $h$ distinct points $s_1,\dots,s_k$ and $t_a$ ($a\in S^c$); it has $k+j=h$ elements. The free–free, free–removed and removed–removed terms assemble to $\sum_{x\neq y\in X}-\log|x-y|$, and

$$\begin{aligned} h^2H_S = {} & \underbrace{\sum_{x\neq y\in X} -\log|x-y|}_{[B]} + \underbrace{2\sum_{i,\,\text{all }b}\log|s_i-t_b|}_{[R1]} \\ & + \underbrace{\sum_{\text{all } b\neq b'} -\log|t_b-t_{b'}|}_{[R2]} \\ & + \underbrace{2\sum_{a\in S^c}\sum_{b\neq a}\log|t_a-t_b|}_{[R3]} + \underbrace{\sum_{i} -\log w_\kappa(z_i)}_{[R4]}. \end{aligned}$$

For $S=\{0,\dots,h-1\}$ the term $[R3]$ is absent and this is the decomposition in the proof of Lemma 4.8.

**The term $[B]$ (replaces circle smoothing).** The pairs of removed nodes occur in $[B]$ and once more, with the opposite sign, in the second identity above; so *the interaction of the removed nodes among themselves is counted exactly once and no energy of the removed set is left over.* Apply Lemma 4.3 to every ordered pair of distinct points of $X$, free or removed, and then Corollary 4.7 with this $X$:

$$\begin{aligned} [B] = \sum_{x\neq y\in X} -\log|x-y| \;&\geq\; \sum_{x\neq y\in X} k_\rho(x,y) \\ &\geq\; h^2\Big[\tfrac{2}{h}\textstyle\sum_{x\in X}U^\sigma_\rho(x) - I_\rho(\sigma)\Big] + h\log\rho, \end{aligned}$$

that is, $\frac{[B]}{h^2}\geq\frac{2}{h}\sum_{i\leq k}U^\sigma_\rho(s_i)+\frac{2}{h}\sum_{a\in S^c}U^\sigma_\rho(t_a)-I_\rho(\sigma)-\frac{|\log\rho|}{h}$. There is no Lipschitz term, no smoothing and no continuum energy: only the finite-sum Lemma 4.6.

**The terms $[R1]$, $[R2]$, $[R4]$.** As in Lemma 4.8: $[R1]/h^2\geq -\frac{2}{h}\sum_i U^\tau(s_i)-\frac{2k}{h^2}\log(1+\frac{8h}{3})$ (monotonicity of $u\mapsto\log|s-u|$ on $[0,1]$ for $\operatorname{Re}s\leq 0$, and $|s_i|\geq\frac{3}{8h}$), and $[R4]/h^2\geq\frac{2}{h}\sum_i(1-\kappa)\pi|\operatorname{Im}s_i|-\frac{k}{h^2}\log^+C_E$. The term $[R2]=h(h-1)\log h-2\sum_{n<h}\log n!$ involves all $h$ nodes and is independent of $S$. Using $\log n!\leq(n+\frac12)\log n-n+1$ for $n\geq1$ gives $[R2]/h^2\geq\frac32-\varepsilon_2(h)$ with $\varepsilon_2(h)=O((\log h)/h)$. Nodes carry no weight, so no field term arises at the removed nodes.

**The term $[R3]$: the removed-node term.** For a node $a$ with $t = a/h$,

$$\sum_{b\neq a} \log|t_a - t_b| = \log(a!\,(h-1-a)!) - (h-1)\log h.$$

By $n! \geq (n/e)^n$, valid for all $n \geq 0$, applied to $a!$ and to $(h-1-a)! = (h-a)!/(h-a)$, we get $\log(a!(h-1-a)!) \geq h[t\log t + (1-t)\log(1-t)] + h\log h - h - \log h$, that is, $\sum_{b\neq a}\log|t_a - t_b| \geq -h\,U^\tau(t_a)$ *with no error term.* Hence $[R3]/h^2 \geq -\frac{2}{h}\sum_{a\in S^c} U^\tau(t_a)$.

**Collecting.** On the real line $U^\sigma_\rho - U^\tau = g$, so

$$H_S \;\geq\; \frac{2}{h}\sum_{i\leq k} g(s_i) + \frac{2}{h}\sum_{a\in S^c} g(t_a) - I_\rho(\sigma) + \frac{3}{2} - \eta_1(h), \tag{7.3}$$

with $\eta_1(h) = \frac{|\log\rho|}{h} + \frac{2\log(1+8h/3)}{h} + \varepsilon_2(h) + \frac{\log^+ C_E}{h}$, which uses only $k \leq h$ and so is uniform in $S$.

*Removed nodes.* Since $t_a \in [0,1)$, (L) gives $\frac{2}{h}\sum_{a\in S^c} g(t_a) \geq 2\beta\, g_{\min}$. As $g_{\min} < 0$, a removed node costs.

*Free points (Lemma 4.9).* The free points have mass $k/h = 1-\beta$, and Lemma 4.9 gives

$$\frac{1}{h}\sum_{i\leq k} g(s_i) \;\geq\; (1-\beta)m' + S_{\mathrm{cert}} - \frac{2}{h}\sum_{\ell}(m' - \gamma_\ell)^+.$$

Inserting both into (7.3),

$$\begin{aligned} H_S &\geq 2(1-\beta)m' + 2S_{\mathrm{cert}} + 2\beta g_{\min} - I_\rho(\sigma) + \tfrac{3}{2} - \eta_1 - \tfrac{4}{h}\textstyle\sum_\ell (m'-\gamma_\ell)^+ \\ &= D_{\mathrm{cert}} - \beta\cdot 2(m' - g_{\min}) - \eta(h), \end{aligned}$$

where (since $D_{\mathrm{cert}} = 2(m' + S_{\mathrm{cert}}) - I_\rho(\sigma) + \frac{3}{2}$, with $S_{\mathrm{cert}}$ the sum of Lemma 4.9) the expression in the statement is exactly $D_{\mathrm{cert}} - \beta c$, and $\eta = \eta_1 + \frac{4}{h}\sum_\ell (m'-\gamma_\ell)^+$ is independent of $S$. This is (7.2). □

7.4. **The certificate for $g_{\min}$.** The number $g_{\min}$ is part of the certificate of §4.4, produced by the same script `kcert.py` on the same `sigma300.json`. The interval $[-\frac{1}{64}, 1] \supset [0,1]$ is divided into 4423 exact cells and each cell becomes one Arb enclosure [1] (full-cell enclosures, so no Lipschitz bound is needed). On each cell the lower bound for $U^\sigma_\rho$ is the one described in §4.4, and $U^\tau(t) = 1 - t\log|t| - (1-t)\log|1-t|$ has $(U^\tau)' = \log|1-t| - \log|t|$, which is positive for $t < \frac{1}{2}$ and negative for $t > \frac{1}{2}$ (also across 0), so its supremum on a cell $[a,b]$ is $U^\tau(b)$ if $b \leq \frac{1}{2}$, $U^\tau(a)$ if $a \geq \frac{1}{2}$, and $U^\tau(\frac{1}{2}) = 1 + \log 2$ otherwise. The certified infimum is

$$g_{\min} \;\geq\; -1.4640996549,$$

with the minimiser near $t \approx 0.7499$ in floating point. Hence $c = 2(m' - g_{\min}) \leq 2(1.9718000062 + 1.4640996549) = 6.8717993\ldots \leq c_{\mathrm{cert}} := 6.8717994$.

### 7.5. **The window corollary.**

**Corollary 7.2.** *For $h \geq 2^{30}$ and every real $t$ with $|t| \leq e^{-ch}$,*

$$|P_h(\lambda+t)| \;\leq\; \exp\bigl(-D_{\mathrm{cert}}h^2+h^2\eta(h)+h\log\bigl(2(2h+c_\kappa)\bigr)\bigr) = e^{-(D_{\mathrm{cert}}-o(1))h^2},$$

*where the $o(1)$ does not depend on $t$. In particular this holds for $|t| \leq e^{-c_{\mathrm{cert}}h}$.*

*Proof.* By (7.1), grouping the subsets $S$ by $j = h - |S|$ (there are $\binom{h}{j}$ of them) and applying Theorem 7.1, with $\beta = j/h$ so that $e^{h^2\beta c} = e^{jch}$,

$$|P_h(\lambda+t)| \leq (2h+c_\kappa)^h e^{h^2\eta} e^{-D_{\mathrm{cert}}h^2} \sum_{j=0}^{h} \binom{h}{j} 16^{-(h-j)} \bigl(|t|e^{ch}\bigr)^j.$$

(Here $(2h+c_\kappa)^k \leq (2h+c_\kappa)^h$ for $2h+c_\kappa \geq 1$.) If $|t|e^{ch} \leq 1$ every summand is at most $\binom{h}{j}$, and the sum is at most $2^h$. □

*Remark* 7.3 (earlier versions). An earlier version of this paper used a Gram-eigenvalue bound, giving $\mu(\lambda) \leq 126$. The node-removal bound replaced it, first with the circle-smoothing energy step (Remark 4.12, $\mu(\lambda) \leq 79.35$) and now with the $k_\rho$ energy step.

## 8. Proof of the main theorem

**Theorem 8.1.** *Let $\eta = D_{\mathrm{cert}} - \Lambda \geq 0.1304266$ and $c_{\mathrm{cert}} = 6.8717994$. Then for every $\nu > c_{\mathrm{cert}}\,(1/\eta+3/4)$ there are only finitely many rationals $p/q$ with $|\lambda - p/q| < q^{-\nu}$. In particular, since $c_{\mathrm{cert}}(1/\eta+3/4) \leq 57.841$ (it is 57.8410 for the unrounded values) and $57.841 < 57.85$, Theorem 1.2 holds.*

*Proof.* Let $\nu > c_{\mathrm{cert}}(\frac{1}{\eta}+\frac{3}{4})$ and suppose $|\lambda - p/q| =: |t| < q^{-\nu}$ with $q$ large; put $L = \log q$. Choose $\delta > 0$ so small that still $\nu > c_{\mathrm{cert}}\,(1+\delta)(\frac{1}{\eta}+\frac{3}{4})$, and set

$$k = \frac{1+\delta}{\eta}, \qquad X = \tfrac{4}{3}kL, \qquad Y = X + (1+\delta)L.$$

*Choice of $\ell$.* By the prime number theorem in arithmetic progressions mod 4 (de la Vallée Poussin; [3, Ch. 20], [9, Cor. 11.19]), $\vartheta(x;4,3) = x/2 + o(x)$, so the primes $\ell \equiv 3 \pmod 4$ in $(X,Y]$ have product $\exp\bigl(\frac{1}{2}(Y-X)(1+o(1))\bigr) > q^{1/2}$ for large $q$. The primes $\ell$ with $v_\ell(q) = 2$ have product at most $q^{1/2}$, since their squares divide $q$. Hence there is a prime $\ell \equiv 3 \pmod 4$ in $(X,Y]$ (so $\ell \geq 7$) with $v_\ell(q) \neq 2$.

*Choice of $h$.* Put $K = (\ell+1)/4$. If $\ell \nmid q$, Theorem 6.2 gives $h \in \{\ell - K, \ell - K - 1\} = \{\frac{3\ell-1}{4}, \frac{3\ell-5}{4}\}$ with $P_h(p/q) \neq 0$. If $\ell \mid q$, then $v_\ell(q) = 1$ or $\geq 3$ and Lemma 5.5 gives $P_h(p/q) \neq 0$ for every admissible $h$, in particular for $h = \ell - K$. In every case

$$\tfrac{3\ell-5}{4} \leq h \leq \tfrac{3\ell-1}{4}, \qquad \text{so} \qquad kL - \tfrac{5}{4} < h < \tfrac{3}{4}Y$$

(because $\ell > X = \frac{4}{3}kL$ and $\ell \leq Y$), $h \geq 2^{30}$ for large $q$, and $Q_h(p/q) \neq 0$, whence by Lemma 2.4

$$|Q_h(p/q)| \geq q^{-h} = e^{-hL}.$$

*Window.* Here $\frac{3}{4}Y = kL + \frac{3}{4}(1+\delta)L = (1+\delta)(\frac{1}{\eta} + \frac{3}{4})L$, so $h < \frac{3}{4}Y$ and the choice of $\delta$ give $c_{\mathrm{cert}}\, h < \nu L$. Hence $|t| < e^{-\nu L} \le e^{-c_{\mathrm{cert}} h} \le e^{-ch}$, and Corollary 7.2 and Theorem 3.1 give

$$|Q_h(p/q)| \le \mathrm{den}(P_h)\,|P_h(\lambda+t)| \le \exp\big((\Lambda - D_{\mathrm{cert}} + o(1))h^2\big) = \exp\big(-(\eta - o(1))h^2\big).$$

*Contradiction.* Combining, $(\eta - o(1))h \le L$. But $h > kL - \frac{5}{4}$ gives $\eta h \ge (1+\delta)L - \frac{5}{4}\eta$, a contradiction for large $q$. □

*Remark* 8.2 (sensitivity). The exponent is $c(\frac{1}{\eta} + \frac{3}{4})$ with $\eta = D_{\mathrm{cert}} - \Lambda$ and $c = 2(m' - g_{\min})$. The term $c/\eta = 52.687$ is what remains if the interval for $\ell$ has length 0; the term $\frac{3}{4}c$ comes from the length $(1+\delta)L$ of that interval, which is needed to find a prime $\ell \equiv 3 \pmod 4$ with $v_\ell(q) \ne 2$, multiplied by $h/\ell \approx \frac{3}{4}$. Lemma 5.5 is what lets one use $\ell \mid q$ when $v_\ell(q) \ne 2$; without it one must avoid every $\ell \mid q$, the interval for $\ell$ must have length $(2+\delta)L$, the term $\frac{3}{4}$ becomes $\frac{3}{2}$, and the bound is $c(\frac{1}{\eta} + \frac{3}{2}) = 62.995$ (see the table of §4.4).

*Remark* 8.3 (a simpler choice of $h$ gives 77.13). If in the proof of Theorem 8.1 one uses Theorem 5.4 instead of Theorem 6.2, that is $h \in \{\ell - 1, \ell - 2, \frac{3\ell-1}{4}\}$ (and $h = \ell - 1$ when $\ell \mid q$), only $\frac{3\ell-1}{4} \le h \le \ell - 1$ is available. With $X = \frac{4}{3}kL$, $Y = X + (1+\delta)L$ as above one gets $kL - \frac{1}{4} \le h \le Y$, and the window condition $c_{\mathrm{cert}}\, Y < \nu L$ requires $\nu > c_{\mathrm{cert}}(\frac{4(1+\delta)}{3\eta} + 1 + \delta)$. The rest of the proof is unchanged and gives: for every $\nu > c_{\mathrm{cert}}\,(4/(3\eta) + 1)$ there are only finitely many $p/q$ with $|\lambda - p/q| < q^{-\nu}$. Since $4/(3\eta) + 1 \le 11.2229$ ($11.22286\ldots$ unrounded) and $c_{\mathrm{cert}} \cdot 11.2229 \le 77.1217 < 77.13$ (unrounded 77.1212), this gives $\mu(\lambda) \le 77.13$. The factor $\frac{4}{3}$ comes from the spread of the three values of $h$, and without Lemma 5.5 the bound is $c(4/(3\eta) + 2) < 84$ (83.99). The Lean formalization currently covers this version (§10), and it does not use Theorem A.

## 9. The mirror constant

Put

$$\lambda' := \sum_{k \ge 1}\Big(\frac{1}{(4k-3)^2} - \frac{1}{16k^2}\Big) = \Big(\frac{\pi^2}{16} + \frac{G}{2}\Big) - \frac{\pi^2}{96} = \frac{G}{2} + \frac{5\pi^2}{96} = 0.972\ldots,$$

using $\sum_{n \equiv 1(4)} n^{-2} = \frac{\pi^2}{16} + \frac{G}{2}$ (the companion of the sums in §2); thus $G + 5\pi^2/48 = 2\lambda'$ and $\lambda + \lambda' = G$. The proof of Theorem 1.3 is the proof of Theorems 1.1 and 1.2 for the *mirror frame*

$$\bar{F}(z) = \psi(z+1) - \psi(z + \tfrac{1}{4}) = \sum_{j \ge 0}\Big[\frac{1}{z + j + \frac{1}{4}} - \frac{1}{z + j + 1}\Big],$$

with poles $+1$ at $-(j + \frac{1}{4})$ and $-1$ at $-(j+1)$. The shift $\frac{3}{4} \to \frac{1}{4}$ exchanges the roles of the residue classes 1 and 3 modulo 4. We list *every* place where the argument changes; everything not listed is used verbatim. The constants

$\Lambda = 313/105$, $D_{\mathrm{cert}} \geq 3.1113790$, $c_{\mathrm{cert}} = 6.8717994$, $\eta \geq 0.1304266$ are the same, so the numerical value of the exponent is the same, $57.841 \leq 57.85$ (and 77.1213 for the simpler choice of $h$).

9.1. **The matrices.** From $\bar{F}(k) - \bar{F}(k-1) = \frac{1}{k} - \frac{4}{4k-3} = -\frac{3}{k(4k-3)}$ and the analogous identity for $\bar{F}'$ one gets, exactly as for (2.2), (2.3),

$$\bar{L}[a][b] = -\frac{3}{b-a} \sum_{a<k\leq b} \frac{1}{k(4k-3)} \ (a<b), \qquad \bar{L}[a][a] = \bar{F}'(0) + \sum_{k\leq a} \Big( \frac{16}{(4k-3)^2} - \frac{1}{k^2} \Big),$$

and $\bar{F}'(0) = \psi'(1) - \psi'(\frac{1}{4}) = \frac{\pi^2}{6} - (\pi^2 + 8G) = -16\lambda'$. Let $N_0 = N_0^{(h)}$ be the symmetric rational matrix with

$$N_0[a][b] = \frac{3}{16(b-a)} \sum_{a<k\leq b} \frac{1}{k(4k-3)} \ (a<b), \qquad N_0[a][a] = \frac{1}{16} \sum_{k\leq a} \Big( \frac{1}{k^2} - \frac{16}{(4k-3)^2} \Big),$$

and put $\bar{T}_h = \frac{1}{16}\bar{L} = -(\lambda' I + N_0)$, $\bar{P}_h(X) = \det(XI + N_0) \in \mathbb{Q}[X]$ (monic of degree $h$). Then $\bar{P}_h(\lambda' + t) = (-1)^h \det(\bar{T}_h - tI)$, and Lemma 2.4 holds verbatim for $\bar{Q}_h = \mathrm{den}(\bar{P}_h)\bar{P}_h$.

9.2. **Content.** For $p \geq 5$ and $q = p^e$ the least $k \geq 1$ with $q \mid k(4k-3)$ is $k_q = (j_q q + 3)/4$, where now $j_q = 1$ if $q \equiv 1 \pmod 4$ and $j_q = 3$ if $q \equiv 3 \pmod 4$ (indeed $4k - 3 = jq$ forces $jq \equiv 1 \pmod 4$; $k$ and $4k-3$ are coprime for $p \geq 5$, and $k_q < q$). So the classes are *swapped* relative to the construction before Lemma 3.3; the dual certificate (the potentials $u, v$ and Lemma 3.3) is verbatim. Since $|k_q - j_q q/4| \leq \frac{3}{4}$, the per-level bound is $\delta(q) \leq h\, g_{j_q}(q/h) + \frac{3}{2}$, and the error costs $O(h)$ in total. The two integrals $\int g_1 = 21/5$ and $\int g_3 = 37/21$ are now attached to the classes 1 and 3 mod 4 respectively; by the prime number theorem in progressions the two classes have equal weight, so $\Lambda = \frac{1}{2}(\frac{21}{5} + \frac{37}{21}) = \frac{313}{105}$ again. The primes $2, 3$ contribute $O(h \log h)$ as before. Hence $\log \mathrm{den}\, \bar{P}_h \leq (\frac{313}{105} + o(1))h^2$.

9.3. **Nonvanishing.** Let $\ell \equiv 1 \pmod 4$ be prime, $\ell \geq 13$, $K = (\ell+3)/4$ (so $4K - 3 = \ell$), and $K + 1 \leq h \leq \ell - 1$; low and high indices are as in §5.

**Lemma 9.1** ($\ell$-singular structure, mirror)**.** *Among $k$, $4k-3$, $b-a$, 16 and the numerator 3 occurring in the entries of $N_0$ only $4k-3$ can be divisible by $\ell$, and this occurs only for $k = K$. Hence every high diagonal entry of $N_0$ contains the term $-\ell^{-2}$, every low–high entry contains $3/(16K\ell(b-a))$, and all other entries lie in $\mathbb{Z}_{(\ell)}$.*

*Proof.* If $\ell \mid 4k-3$ with $4k-3 \leq 4h-7 < 4\ell$, then $4k-3 \in \{\ell, 2\ell, 3\ell\}$; $2\ell$ is even and $3\ell \equiv 3 \not\equiv 1 \pmod 4$. The numerator 3 is a unit since $\ell \neq 3$. The rest is as in Lemma 5.1. □

**Lemma 9.2.** *With $D = \mathrm{diag}(I_K, \ell I_{h-K})$, modulo $\ell$*

$$D(XI + N_0)D \equiv \begin{pmatrix} XI_K + A & \hat{B} \\ \hat{B}^\top & -I \end{pmatrix}, \qquad \hat{B}_{ab} \equiv \frac{3}{16K(b-a)} \equiv \frac{1}{4(b-a)},$$

*because* $16K = 4\ell + 12 \equiv 12$. *The Schur complement of the block* $-I$ *gives* $\ell^{2(h-K)}\bar{P}_h(X) \equiv (-1)^{h-K}\det(XI_K + A + \hat{B}\hat{B}^\top)$, *which is monic of degree* $K$ *up to sign; so* $v_\ell(\operatorname{den}\bar{P}_h) = 2(h-K)$. *(The sign in front of* $\hat{B}\hat{B}^\top$ *is the* opposite *of that in Lemma 5.2, because the high block is* $-I$ *instead of* $+I$*.)*

**Lemma 9.3** (key identity, mirror)**.** *Let* $s = \ell - h$, $m = \ell - K$, *let* $C_s$ *be as in Lemma 5.3 (note* $a + j \le K - 1 + s < \ell$*), and put* $Y = -(16X + H_m^{(2)})$. *Then in* $\mathbb{F}_\ell[X]$

$$16^K \ell^{2(h-K)}\bar{P}_h(X) \equiv (-1)^h \det(YI_K + C_sC_s^\top).$$

*Proof.* Modulo $\ell$, $4K \equiv 3$, so $\frac{4}{4k-3} = \frac{1}{k-3/4} \equiv \frac{1}{k-K} \equiv \frac{1}{k+m}$, hence $\frac{3}{k(4k-3)} \equiv \frac{1}{k+m} - \frac{1}{k}$ and $\frac{16}{(4k-3)^2} \equiv \frac{1}{(k+m)^2}$. For $a < b < K$ (so $b + m < \ell$)

$$16A_{ab} \equiv -\frac{1}{b-a}\sum_{k=a+1}^{b}\Big(\frac{1}{k} - \frac{1}{k+m}\Big) = -\sum_{j=1}^{m}\frac{1}{(a+j)(b+j)}, \qquad 16A_{aa} \equiv \sum_{k\le a}\Big(\frac{1}{k^2} - \frac{1}{(k+m)^2}\Big) = H_m^{(2)} - \sum_{j}$$

that is $16A \equiv H_m^{(2)} I - C_mC_m^\top$. As in the proof of Lemma 5.3 (substitute $j = \ell - b$, so $b - a \equiv -(a + j)$), $16\hat{B}\hat{B}^\top \equiv \sum_{j=s+1}^{m} w_j w_j^\top = C_mC_m^\top - C_sC_s^\top$. Therefore $16(A + \hat{B}\hat{B}^\top) \equiv H_m^{(2)} I - C_sC_s^\top$, and $\det(XI + \frac{1}{16}(H_m^{(2)} I - C_sC_s^\top)) = 16^{-K}(-1)^K \det(YI + C_sC_s^\top)$. Combine with Lemma 9.2: the sign is $(-1)^{h-K}(-1)^K = (-1)^h$. □

**Theorem 9.4** (choice of $h$, mirror)**.** *Let* $\ell \equiv 1 \pmod 4$ *be prime,* $\ell \ge 13$, $K = (\ell+3)/4$, *and* $r \in \mathbb{Q}$ *with* $\ell \nmid \operatorname{den} r$. *Then one of* $h \in \{\ell - K,\ \ell - K - 1\} = \{\frac{3\ell-3}{4}, \frac{3\ell-7}{4}\}$ *satisfies* $v_\ell(\bar{P}_h(r)) = -2(h-K)$*; in particular* $\bar{P}_h(r) \ne 0$.

*Proof.* Here $s = \ell - h \in \{K, K+1\}$; both values satisfy $K + 1 \le h \le \ell - 1$ and $1 \le s \le \ell - K - 1$ for $\ell \ge 13$, so Lemmas 9.2, 9.3 apply: $\ell^{2(h-K)}\bar{P}_h(r)$ is an $\ell$-adic unit iff $D_s(Y) = \det(YI_K + N_s)$ does not vanish at $Y = y := -(16r + H_m^{(2)}) \in \mathbb{F}_\ell$. Suppose $D_K(y) = D_{K+1}(y) = 0$. As before $D_K(0) = \det(C_K)^2 \ne 0$ (a Cauchy determinant in integers $\le 2K - 1 < \ell$), so $y \ne 0$ and $-y \ne 0$ is a common eigenvalue of $N_K$ and $N_{K+1}$. Lemma CC holds for any $K$, any $s \ge K$ and any prime $\ell > K + s$: its proof (Krylov reduction and Theorem A at $\theta = 0$) uses only that $u(K, s)$ is a rational number all of whose prime factors are integers in $[1, K + s]$, and never mentions the residue class of $\ell$ (Theorem A is an identity in $(n, s, \theta)$). Here $K + s = 2K = (\ell+3)/2 < \ell$ for $s = K$. This contradicts Lemma CC. □

(The same argument gives the three-$h$ version, $h \in \{\ell - 1, \ell - 2, \ell - K\}$, with $\Xi_{11} = H_K^{(2)}$ and $\Xi_{12} = K/(K+1) \ne 0$; this is the route to 77.13.)

**Lemma 9.5** (primes dividing the denominator, mirror)**.** *Let* $\ell \equiv 1 \pmod 4$ *be prime,* $\ell \ge 13$, $K + 1 \le h \le \ell - 1$, $r = p/q$ *with* $e = v_\ell(q) \ge 1$ *and* $u = \ell^e r$. *If* $e = 1$, *or* $e \ge 3$, *or* $e = 2$ *and* $u \not\equiv +1 \pmod \ell$, *then* $\bar{P}_h(r) \ne 0$ *for every admissible* $h$.

*Proof.* As for Lemma 5.5, with high block $\ell^2 rI + J$ and now $J \equiv -I$ (Lemma 9.1): the reduced determinants are $u^K$ for $e = 1$, $u^K(u-1)^{h-K}$ for $e = 2$, and $u^h$ for $e \geq 3$. □

Thus the exceptional case is $e = 2$, $u \equiv +1$ (instead of $u \equiv -1$); since it still requires $v_\ell(q) = 2$, the selection "$v_\ell(q) \neq 2$" in the proof of Theorem 8.1 is unchanged. (Checked on 574 exact cases for $\ell = 13, 17, 29$, all admissible $h$ and $e \leq 4$; the exception $e = 2$, $u \equiv 1$ does occur. Lemma 9.3 was checked by exact characteristic polynomials for $h = 12, 20$ and all $\ell \equiv 1 \pmod 4$ in $(h, 4h]$.)

9.4. **Decay and node removal.** *The certificate is the same.* The certificates for $D_{\mathrm{cert}}$ (§4.4) and for $g_{\min}$ (§7.4) concern one fixed function $g$ on the scaled plane; they do not depend on $F$, on $h$, or on where the atoms lie modulo 1. What changes is only the contour identity and the $o(1)$ bookkeeping, as follows.

*Reflection.* From $\psi(1+z) = \psi(-z) - \pi\cot\pi z$ and $\psi(\frac{1}{4}+z) = \psi(\frac{3}{4}-z) - \pi\cot(\pi z + \frac{\pi}{4})$,

$$\bar{F} = \tilde{F} + \bar{E}, \qquad \tilde{F}(z) = \psi(-z) - \psi(\tfrac{3}{4} - z), \quad \bar{E}(z) = \frac{-\pi\sin(\pi/4)}{\sin\pi z\,\sin(\pi z + \pi/4)},$$

with $\tilde{F}$ analytic and $O(1/|z|)$ in $\operatorname{Re} z < 0$, and $\bar{E}$ 1-periodic with $|\bar{E}(x+iy)| \leq Ce^{-2\pi|y|}$ for $|y| \geq 1$.

*The contour.* Take $N_0 = \lfloor r_0 h \rfloor$ as before and $x^* = -N_0 + \frac{3}{8}$, the midpoint of the gap $(-N_0, -N_0 + \frac{3}{4})$ between consecutive poles of $\bar{F}$, so $x^* \equiv \frac{3}{8} \pmod 1$. On the part of $\Gamma_h = x^* + u(-1 \pm 3i)$ with $|\operatorname{Im} z| < 1$ one has $\operatorname{Re} z \in (-N_0 + \frac{1}{24}, -N_0 + \frac{3}{8}]$, inside that gap. The poles to the right of $x^*$ are $-(j+\frac{1}{4})$, $0 \leq j < N_0$ (residue $+1$), and $-(j+1)$, $0 \leq j < N_0 - 1$ (residue $-1$); so the set of atoms is

$$\bar{\mathcal{A}} = \{j + \tfrac{1}{4} : \ 0 \leq j < N_0\} \ \cup \ \{j + 1 : \ 0 \leq j < N_0 - 1\}, \qquad |\bar{\mathcal{A}}| = 2N_0 - 1,$$

with $c_\mu = +1$ on the first part and $-1$ on the second, and Proposition 4.1 holds verbatim: $\bar{L} = -\sum_{\mu\in\bar{\mathcal{A}}} c_\mu u_\mu u_\mu^\top + \frac{1}{2\pi i}\int_{\Gamma_h} \bar{E}\, vv^\top dz$ (checked entrywise to $2.4 \cdot 10^{-17}$ at 25 digits for $N_0 = 1, 2, 7$, and independently for $(N_0, h) = (4, 3), (3, 4)$).

*Changes in Andréief's bound, Lemmas 4.8, 4.9 and Theorem 7.1.* All are in $o(1)$ terms:

- the support of the measure has $\operatorname{Re} z \leq -\frac{1}{4}$ (the nearest atom is $-\frac{1}{4}$), so $|v_a| \leq 4$ (instead of $\frac{8}{3}$), and the corresponding loss is $\log(1+4h)/h \to 0$;
- $C_E$ and $c_\kappa$ are independent of $h$, by periodicity and $x^* \equiv \frac{3}{8} \pmod 1$ (the contour crosses the real axis at distance $\frac{3}{8}$ from the poles at 0 and $\frac{3}{4}$ modulo 1);
- in Lemma 4.9, $x^*/h = -r_0 + \delta_h$ with $\delta_h = (r_0 h - N_0 + \frac{3}{8})/h \in [\frac{3}{8h}, \frac{11}{8h})$, so $|\delta_h| \leq \omega = 2^{-30}$ as soon as $h \geq \frac{11}{8} \cdot 2^{30}$; the atoms lie in $-(\mathbb{Z}_{\geq 0} +$

$\frac{1}{4})/h \cup -\mathbb{Z}_{\geq 1}/h \subset [-r_0, 0]$ (the largest is $N_0 - \frac{3}{4} \leq r_0 h$), two progressions of spacing $1/h$, so $N_\ell \leq 2h(b_\ell - a_\ell) + 2$ exactly as before.

The function $g$, the boxes (R), the segment cells (S), the line cells (L) and $I_\rho(\sigma)$ are identical. Hence Theorem 4.10 holds for $\bar{P}_h$ with $D^+ = D_{\mathrm{cert}} \geq 3.1113790$, Theorem 7.1 with $c \leq 6.8717994$, and Corollary 7.2: $|\bar{P}_h(\lambda' + t)| \leq e^{-(D_{\mathrm{cert}} - o(1))h^2}$ for $|t| \leq e^{-ch}$.

*Remark* 9.6. Exact determinants (Arb, rational entries) give $-\log|\det \bar{T}_h|/h^2 =$ 3.030, 3.060, 3.075, 3.084, 3.090, 3.094 at $h = 32, 48, 64, 80, 96, 112$, against 3.068, 3.084, 3.093, 3.098, 3.101, 3.104 for $T_h$. The difference of the logarithms is $\approx 1.06\,h + 4.7$, linear in $h$ (the near atom $-\frac{1}{4}$ costs $O(h)$, not $O(h^2)$), so the $h^2$-constants coincide; a margin "$\eta \approx 0.10$" read off from the finite-$h$ slopes is a finite-$h$ artefact. These computations are consistency checks and are not used in the proof.

9.5. **The measure.** The proof of Theorem 8.1 is verbatim with: primes $\ell \equiv 1 \pmod 4$ in $(X, Y]$ (using $\vartheta(x; 4, 1) = x/2 + o(x)$); Theorem 9.4 in place of Theorem 6.2; Lemma 9.5 in place of Lemma 5.5; and $h \in \{\frac{3\ell-7}{4}, \frac{3\ell-3}{4}\}$, so that $kL - \frac{7}{4} < h < \frac{3}{4}Y$. With $\eta$, $c_{\mathrm{cert}}$ as above this yields: for every $\nu > c_{\mathrm{cert}}(1/\eta + \frac{3}{4})$ there are only finitely many $p/q$ with $|\lambda' - p/q| < q^{-\nu}$, and $c_{\mathrm{cert}}(1/\eta + \frac{3}{4}) \leq 57.841 < 57.85$ (57.8409 unrounded). With the three values of $h$ one gets $c_{\mathrm{cert}}(4/(3\eta) + 1) \leq 77.1217$. This proves Theorem 1.3: $\lambda'$ is irrational and $\mu(\lambda') \leq 57.85$; since $\mu(a\xi + b) = \mu(\xi)$ for rational $a \neq 0$ and $b$, also $\mu(G + 5\pi^2/48) \leq 57.85$ and $G + 5\pi^2/48 \notin \mathbb{Q}$.

*Remark* 9.7 (the two measures do not combine). Since $\lambda + \lambda' = G$ one might hope to deduce something about $G$ from Theorems 1.2 and 1.3. This is impossible by any black-box argument. Fix a Liouville number $\theta$. For Lebesgue-almost every real $x$ one has $\mu(x) = 2$ and $\mu(\theta - x) = 2$ (the exceptional sets are null and invariant under $x \mapsto \theta - x$), yet $x + (\theta - x) = \theta$ has $\mu = \infty$. So bounds on $\mu(\xi_1)$ and $\mu(\xi_2)$ say nothing about $\mu(\xi_1 + \xi_2)$, and the pair of results gives no measure for $G$ (nor a new proof that $G$ is irrational, which is OpenAI's theorem [10]). Theorems 1.2 and 1.3 concern two different linear forms, $G - 5\pi^2/48$ and $G + 5\pi^2/48$, in $1, G, \pi^2$.

## 10. Formalization status

We state this carefully. The companion paper [6] on Catalańs constant is fully formalized in Lean 4, and so are the irrationality of $\lambda$ and the bound $\mu(\lambda) \leq 77.28$ proved here; Theorem A, Lemma CC and the bound 57.85 are not. Each of the Lean results below compiles without `sorry` and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The library uses the Lean module system; a full build of the $\lambda$ library takes about 14 minutes, and the kernel-checked numerical certificate about one CPU-hour.

- **Formalized (Lean 4 and Mathlib): nonvanishing.** The $\ell$-adic nonvanishing theorem, Theorem 5.4: for a prime $\ell \equiv 3 \pmod 4$, $\ell \geq 7$, and $r \in \mathbb{Q}$ with $\ell \nmid \operatorname{den} r$, one of $P_{\ell-1}(r), P_{\ell-2}(r), P_{(3\ell-1)/4}(r)$ is nonzero. The theorem is `CatalanLambda.lambda_nonvanishing`. The formalized proof is organized slightly differently from §5: it works in the subring $\mathbb{Z}_{(\ell)}$ with the reduction map to $\mathbb{Z}/\ell$, uses the telescoping $\frac{1}{k} - \frac{1}{k+\ell-K} = -\frac{1}{k(4k-1)}$ directly, and absorbs the constant $-H_m^{(2)}$ into a single constant. It does not formalize the valuation statement $v_\ell = -2(h-K)$.
- **Formalized: the prime number theorem in arithmetic progressions.** The theorem for every modulus, used in Theorems 3.1 and 8.1, as `CatalanLambda.pnt_AP`, together with the upper bound `CatalanLambda.theta_mod4_upper`. It is derived from the vendored Wiener–Ikehara theorem and Mathlib's results on Dirichlet $L$-functions.
- **Formalized: the content bound.** The integrality estimate (Lemma 2.4) as `CatalanLambda.integrality`, and the denominator bound of Theorem 3.1 in the form `CatalanLambda.log_Dden_le_eventually`.
- **Formalized: the Löwner form.** The off-diagonal and diagonal formulas for the matrices $T_h$ as Löwner matrices (Proposition 2.2), as `CatalanLambda.Decay.lowner_offdiag` and `CatalanLambda.Decay.lowner_diag`, and the identity $\lambda = G/2 - 5\pi^2/96$ as `CatalanLambda.Decay.lamS_eq`.
- **Formalized: the contour identity.** Proposition 4.1 as `CatalanLambda.Decay.contour_identity`
- **Formalized: Andréief and Cauchy.** The Andréief-type determinant bound (Proposition 4.2) as `CatalanLambda.Decay.tent_andreief_bound`, and the Cauchy determinant as `CatalanLambda.Decay.cauchy_det`.
- **Formalized: the kernel $k_\rho$, Theorem 7.1 and the deduction.** The positive-definiteness lemma for the kernel $k_\rho$ (Lemmas 4.4, 4.5, 4.6 and Corollary 4.7); the contour identity for a general ray angle (including the direction $(-1, 3)$ used here); Theorem 7.1 (node removal $N_\rho$), as `CatalanLambda.Decay.theoremN`, together with `CatalanLambda.Decay.window_liminf_P`; Lemma 5.5 as `CatalanLambda.Measure.lemma3_pq`; and the deduction of the irrationality and of the bound of Remark 8.3 from the certificate facts, as `CatalanLambda.Measure.lambda_irrational` and `CatalanLambda.Measure.lambda_measure_one`.
- **Formalized: the numerical certificate.** The structure `CatalanLambda.Cert.certFacts` collecting the numerical facts of §4.4 and §7.4 (the certified values of $D_{\mathrm{cert}}$, $m'$, $S_{\mathrm{cert}}$, $g_{\min}$) is inhabited by a kernel-checked proof: about 27.6k ray boxes, 2.8k segment cells and 512 line cells, the bound on $I_\rho$ and the tail estimate. The proof uses a far-field first-order Taylor model with verified rational tables for the logarithm and the arctangent, and takes about one CPU-hour in the kernel. Lean works with the short rational constants $D_{\mathrm{cert}} = 31111/10000$ and $c_{\mathrm{cert}} = 68726/10000$ obtained by rounding the certified values, which is why it proves 77.28 rather than 77.13.

- **Formalized: the final theorems.** `CatalanLambda.lambda_irrational_final` ($\lambda$ is irrational), `CatalanLambda.catalan_sub_irrational` ($G - 5\pi^2/48$ is irrational), `CatalanLambda.lambda_measure_final` ($\mu(\lambda) \le 31682445459/409965000 \approx 77.28$) and `CatalanLambda.lambda_measure_78_catalan` (the round exponent 78), all unconditional. The statements are in `ChallengeLambda.lean`, which needs only Mathlib and writes $G$ as the series $\sum_{j\ge0}(-1)^j/(2j+1)^2$; the proofs are in `SolutionLambda.lean`.
- **Formalized: the mirror theorem (weaker exponent).** The irrationality of $\lambda' = G/2 + 5\pi^2/96$ and of $G + 5\pi^2/48$, and the measure with the same exponent $31682445459/409965000 \approx 77.28$ (hence 78; and 79 for $G + 5\pi^2/48$) are formalized in `CatalanLambda/Mirror/`, with the three-$h$ nonvanishing for $\ell \equiv 1 \pmod 4$ (no Theorem A), the mirror contour identity, and the *same* kernel-checked certificate `CatalanLambda.Cert.certFacts`. The final statements are `CatalanLambda.Mirror.lambda_mirror_measure_78_catalan`, `CatalanLambda.Mirror.catalan_` and `CatalanLambda.Mirror.catalan_add_measure_79`, with challenge statements `CatalanLambdaChallenge.lambdaMirror_irrationalityMeasure_78` and `CatalanLambdaChallenge.catalan_add_irrationalityMeasure_79`; axioms `propext`, `Classical.choice`, `Quot.sound`. The bound 57.85 for $\lambda'$ (via Theorem A) is not formalized.
- **Not formalized.** Theorem A, Lemma CC and Theorem 6.2 (§6), and the deduction of the bound 57.85 in Theorem 8.1. The Lean statement is the weaker bound 77.28, obtained with the simpler choice of $h$ (Remark 8.3) and the rounded constants.
- **Computer-assisted.** Theorem 4.10 and Theorem 7.1 were established by interval-arithmetic computations in Arb (§4.4 and §7.4), carried out by the Python script `kcert.py` on the certificate measure `sigma300.json` (produced from a float optimum by the dyadic rounding script `roundsig.py`; `mkdata.py` re-derives the data in exact rationals; `scan.py` is the supplementary content scan). The script was re-run by an independent audit agent, which reproduced the numbers of §4.4 and sampled the certified cell bounds against floating-point values at about $5 \times 10^5$ ray points, $4 \times 10^4$ segment points and $10^4$ points of the line; but this sampling is not a proof; the certificate facts themselves are re-proved in Lean (the item on the numerical certificate above), by a separate kernel-checked computation. The float construction of $\sigma$ need not be correct since the certificate is checked as it stands. The scripts are archived in the directory extttpaper-lambda/scripts/ of https://github.com/slitsey/catalan-irrationality-measure.

The Lean sources are in the directory `CatalanLambda/` of the repository https://github.com/slitsey/catalan-irrationality-measure, with the files `ChallengeLambda.lean` and `SolutionLambda.lean` at its top level.

All other claims were audited by independent Claude (Anthropic) agents who re-derived the statements by hand and re-ran the numerical checks: the content bound; Lemmas 5.2, 5.3 and Theorem 5.4 (exact characteristic polynomials at $h = 8, 12, 16, 20$ and all admissible $\ell$; gcd $= 1$ for $\ell = 7, \dots, 383$; no violation of the tropical bound against the exact $v_p(\operatorname{den} P_h)$ at $h = 32, 64$); Lemmas 4.4, 4.5, 4.6 and Corollary 4.7 (by hand, with a random numerical test of the inequality in 2000 trials); the contour identity for the direction $(-1, 3)$ (by hand and to $10^{-17}$ numerically); Lemma 4.9 and Theorem 7.1, checked line by line against the circle-smoothing version; the certificate re-run (reproducing $D_{\mathrm{cert}} \geq 3.11137905$, $c \leq 6.87179932$); Lemma 5.5 (1044 exact cases); and the simpler deduction of Remark 8.3 ($4/(3\eta) + 1 = 11.2228582$, $77.12123 < 77.13$). The new material of §6 and Theorem 8.1 was audited by a further independent Claude agent (full pass): Theorem A (including the sign) was checked in exact rational arithmetic at 90 random rational pairs $(\theta, s)$, including non-integer and negative $s$, for $n \leq 10$, and against the direct Cauchy Gram matrices ($\theta = 0$, $s \leq 2n + 1$, $n \leq 10$); $u(K, K)$ modulo $\ell$ is nonzero and equals the closed form for all 86 primes $\ell \equiv 3 \pmod 4$ with $7 \leq \ell < 1000$; the resultant identity (R) was checked exactly; the characteristic polynomials of $N_K$, $N_{K+1}$ are coprime for 39 primes $\ell < 400$; Lemma 5.3 was checked for $\ell = 7, 11, 19, 23$ and $h = \ell - K, \ell - K - 1$; and the numbers $6.8717994\,(1/0.1304266 + 0.75) = 57.84095$, 62.9948 (without Lemma 5.5) and 77.1213 were recomputed. The formula of Theorem A was also verified exactly for every $n \leq 12$. Exact computation of $u(K, K)$ modulo $\ell$ for all $\ell \equiv 3 \pmod 4$, $7 \leq \ell < 4500$, gave no zero. The deduction of Theorem 8.1 is written out in §8 from the audited arithmetic. The audit listed only cosmetic issues, which are corrected in this text (notably the factor $4/h$ in $\eta$).

The mirror section §9 was audited by a separate Claude agent (full pass, no fatal issues; four cosmetic points, corrected here). Lemmas 9.1–9.3 were re-derived by hand (Schur complement with high block $-I$); $N_0$ was checked against $-(\lambda' I + \bar{T}_h)$ to $10^{-51}$ at $h = 9$; the reflection $\bar{F} = \tilde{F} + \bar{E}$ was checked by hand and to $10^{-22}$, and the contour identity independently to $2 \cdot 10^{-17}$, with the atom set compared to the poles right of $x^*$; the valuation $2(h - K)$ and degree $K$ were checked for all 131 admissible $(\ell, h)$ with $\ell = 13, 17, 29, 37$; the characteristic polynomials of $N_K, N_{K+1}$ are coprime over $\mathbb{F}_\ell$ for all 58 primes $\ell \equiv 1 \pmod 4$, $13 \leq \ell < 700$; Lemma 9.5 deviates only for $e = 2$, $u \equiv 1$; $v_p(\operatorname{den} \bar{P}_h) \leq \sum_e \delta(p^e)$ for all $p \geq 5$ at $h = 16, 32, 64, 128$; and $6.8717994\,(1/0.13042662 + 0.75) = 57.84094$, 77.12126 were recomputed. This is not a substitute for human refereeing.

## Acknowledgments and provenance

The irrationality of Catalan's constant, and the Lean formalization on which the author's earlier work builds, are due to OpenAI [10]. The mathematics of this paper —the matrices $T_h$, the content, decay, nonvanishing

and node-removal arguments, the numerical certificates and the text—were derived by Claude (Anthropic) agents working at the request of the author, who posed the problem, directed the work, and is responsible for this submission; independent Claude agents performed the audits described in §10. The Lean formalization was likewise written by Claude agents working with the author. The repository infrastructure, vendored formalizations and certificate tooling of the companion project, which this work reuses, were developed with OpenAI Codex as described in [6]. We use Lean 4 [8], Mathlib [7], the PNT+ project [11] (whose Wiener–Ikehara theorem we use to formalize the prime number theorem in progressions), and Arb [1] for interval arithmetic. The use of AI in this work is disclosed here in the same spirit as in [6].

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